1. INTRODUCTION

1.1 INTRODUCTION

A helmet-mounted display (HMD) is a device used in aircraft to project information to the pilot's eyes. Its scope is like that of head-up displays (HUD) on an aircrew's visor or reticule. An HMD provides the pilot with situation awareness, an enhanced image of the scene, and in military applications cue weapons systems, to the direction their head is pointing. Applications which allow cuing(targeting) of weapon systems are referred to as helmet-mounted sight and display (HMSD) or helmet-mounted sights (HMS). **UAV**

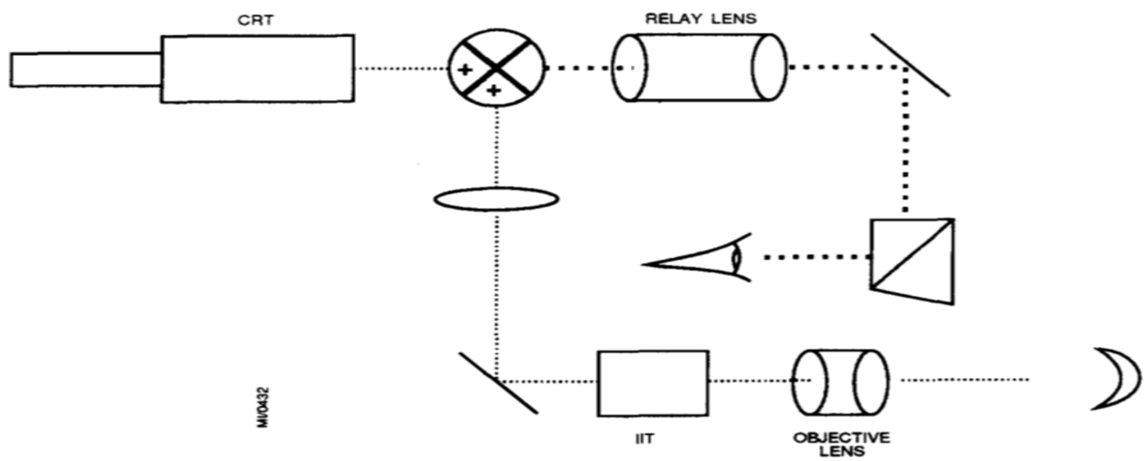

Figure 1.1 optical combination of CRT and intensified imagery

1.2 AIM OF THE PROJECT

To proof the concept of transparent HMD using OLED.

1.3 OBJECTIVES OF THE PROJECT

- Literature Survey on HMD and OLED.
- To design the electronics unit for the display.
- To generate display symbology using bitmapping.
- Choosing of suitable OLED material.
- Conceptual modeling of the HMD

1.4 EXPECTED OUTCOME

- To remove the need for an eyepiece in HMD.
- Reduction in weight of HMD, due to the absence of projecting units and lenses.
- Solving the problem of monocular rivalry in current HMDs.

2. LITERATURE REVIEW

2.1 INTRODUCTION

Several research has been carried out, on the transition from the traditional CRT projected HMD to a lighter weight and efficient HMD. Some of this relevant researches are reviewed below.

2.2 STUDIES ON HELMET OPTICS

The other important issue is that of providing the pilot with a reference reticle to place over the target. The direction of the line between the reticle and the pilot's eye is the line of sight (LOS) between the aircraft and the intended target. The simplest strategy used here is that employed in the US Navy's seventies VTAS (F-4N/AIM-9H), which is a simple mechanical "ring and bead" style sight, attached to the front of the helmet. While it is simple to do, it by its nature will always be out of focus relative to the target, which is optically at infinity. Also, depending on the mounting method, it may move under G load and introduce an unwanted angular error. The next step up is to use a flat optical combiner glass, based on the same principle as a WW2 optical gunsight, and project a simple optically collimated reticle, focussed at infinity. This adds some weight to the helmet, but solves the problem of reticle focus. A light source such as an LED with a suitable lens package mounted in the top of the helmet solves this problem. Just as the optically collimated gunsight evolved into the HUD, with the incandescent bulb replaced by a Cathode Ray Tube (CRT), so the nineties have seen the introduction of embedded CRTs into helmets, thus transforming the HMS into a HMD.Modern HMDs employ typically a compact pencil shaped CRT embedded in the helmet, and suitable optics to focus the CRT image on to the pilot's visor. Focussing at infinity is often performed by matched visor shape (i.e. curvature) and clever optics. Using this arrangement, a field of view between 10 and 30 degrees can be readily accommodated. With a CRT the pilot can be presented with proper symbology, not just a circle or set of crosshairs. Missile status data, seeker acquisition envelope and critical flight data such as load factor, speed, AoA (angle of attack), altitude, Mach number and horizon may be presented. Cueing boxes produced from radar and RWR/ESM data can be used to mark the position of a target at the limits of visual range. More advanced HMDs can also project raster imagery for the pilot, derived from an external steerable FLIR or an NVG image intensifier tube(s) embedded in the helmet. With the availability of Platinum Silicide near infrared and Indium Antimonide mid infrared single chip Focal Plane Array imagers, the long-term outlook is that a HMD can also provide the pilot with night vision over a wide field of view. The most problematic issue at this time with fighter HMDs is weight. The technology exists to embed various imaging sensors in the helmet, but this is not particularly compatible with high G maneuvering. Indeed, this is the reason why elaborate HMDs are becoming common for attack helicopter applications, but have yet to see wider application in fighters.

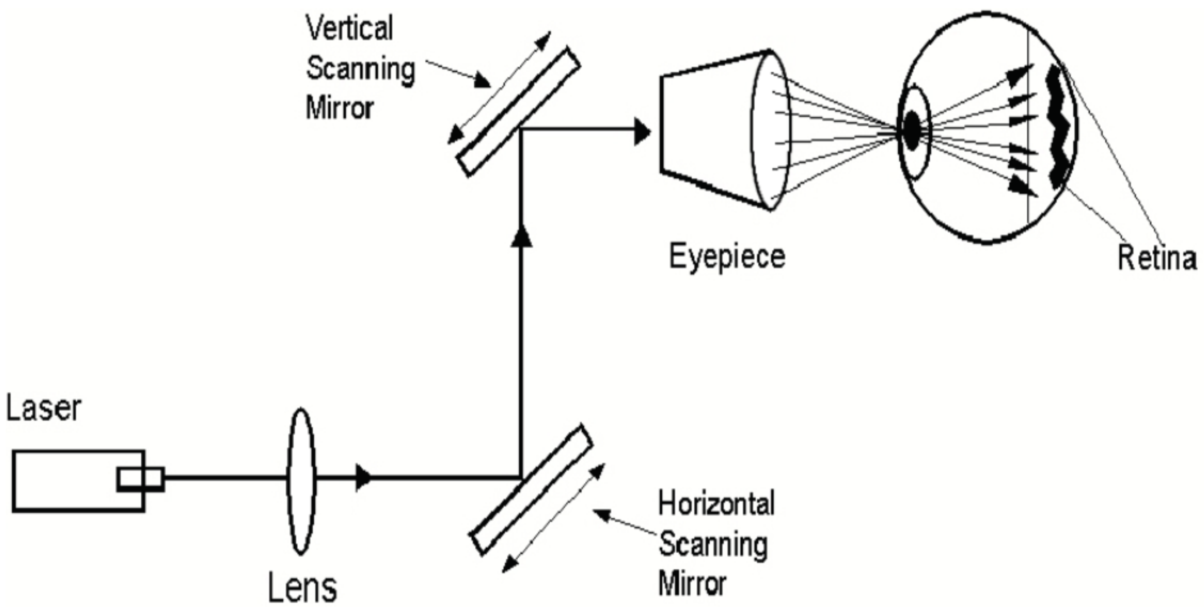

Figure 2.1 Basic diagram of retinal scanning display (adapted from Proctor, 1996).

The other major problematic issue with HMDs is that the user must keep his/her eye always aligned with the sight, and thus the natural combination of head and eye movement used to track a moving object must be unlearned, and a new "head only" tracking motion learned. Since the position of the helmet is what is used to point the missile, pilots must develop even greater neck musculature. The technology is far from its limits now. The virtual reality technology base, much better funded from commercial sources, has produced several interesting ideas which are likely to find their way into fighter helmets. One such idea, which has been demonstrated in the laboratory, is direct retinal projection. In schemes based on this model, a small mirror is mounted in front of the eye.

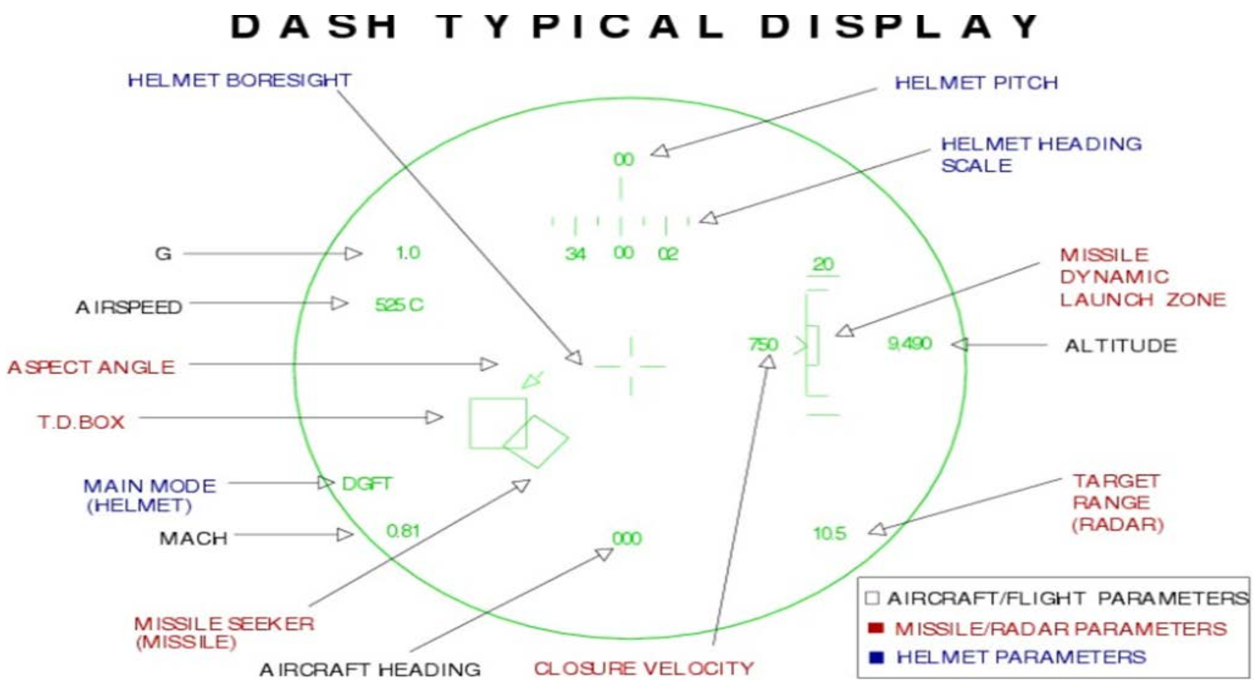

Figure 2.2 Typical dashboard display

A low power laser, or rather a set of three (red, green, blue) lasers are bounced off the mirror directly through the pupil to project a raster scan image directly on the retina. The light beam is scanned by a tiny piezo-electric actuator, which deflects the mirror at a very high speed. While this technology is claimed to provide the potential for image resolutions of 8000 x 4000 pixels, existing technology is limited to that of a VGA class computer monitor.Safety is needless to say an issue, since the failure of the mirror actuator could burn a spot in your retina, indeed existing prototypes have elaborated fail sense mechanisms to shut off the laser beams in the event of deflection system failure. Problematic issues remain with direct retinal projection. One is that the existing head only tracing motion problem is not solved. The other the is potential for eye fatigue due to the retina being driven with an intense light source, and thus requiring recovery time after use. Another idea from the virtual reality world which could prove to be useful for HMD designs is eye tracking, or sensing the direction the eyeball is looking at, relative to the direction of the head. A typical eye tracker will combine a miniature CCD camera and an infrared LED to illuminate the pupil. By measuring the changes in shape and position of the pupil (i.e. an ellipse changing its proportions and position) it is possible to measure the direction the eye is looking in, with very reasonable accuracy once calibrated. Figures cited are as good as tens of seconds of arc.

Combining measurements of head position and eye position would solve the problems inherent in existing HMD technology, since the symbology and weapon boresight could be slaved to the direction the pilot is looking in, thus retaining existing (instinctually programmed) human tracking behavior. The impediment now to the adoption of eye tracking into existing CRT projection HMDs is the limited field of view of the CRT projection optics. Future HMDs with wider field of view optics will be capable of exploiting eye tracking techniques.

2.3 Studies on HMD Designs

The range of HMD design types run from a simple projection of symbolic and/or alpha/numeric information overlaying a direct-view real image for day time use, to a head slaved virtual imaging device that could be linked to remote sensors and/or computer generated imagery for day or night use, and of course, anything in between. As the design type increases in complexity, so does the optical design.

2.3.1 Simplest HMD Design

In the Vietnam Era, a Bell Cobra helicopter (AH-1) was developed with a simple monocular helmet sight (known as the Cobra sight) that could translate an external mounted machinegun using a mechanical head tracker that attached to the top of the helmet (Braybrook, 1998). In front of the right eye was a small semi-transparent window that projected a red dot that was similar to simple commercial red dot reflex sights on some pistols and rifles. The 17-millimeter (mm) diameter combiner was located outside the helmet visor about 50 mm from the eye and could be adjusted in vertical and horizontal positions to properly align with the right eye. The size of the projected red dot was only a few milliradians (mr) in diameter, and was focused at infinity. The see-through visible transmission of the combining glass (beamsplitter) was very high, and the brightness of the aiming reticule was sufficient to be visible at the sky horizon. Complex modern HMD designs In contrast to the simple design of the Cobra sight is a limited-fielded visor-projection HMD currently being offered by a leading aerospace company having the following characteristics:

- Visor projection optical design
- Focused and aligned at infinity
- Binocular viewing

- Magnetic or electro-optical head tracked
- See-through vision
- FOV – >40°
- Can accommodate a wide range of eye separations
- Sufficient brightness and contrast for both day and night operations
- Incorporates direct-view image intensifiers on the side of the helmet and video links to external sensors reflected from the visor
- Flight and/or systems symbology can be projected, with unaided daytime see-through vision or at night, overlaying the image intensified or thermal image.

2.3.2 Optical Design Parameters

There are several important descriptive parameters in an HMD optical design. These includes:

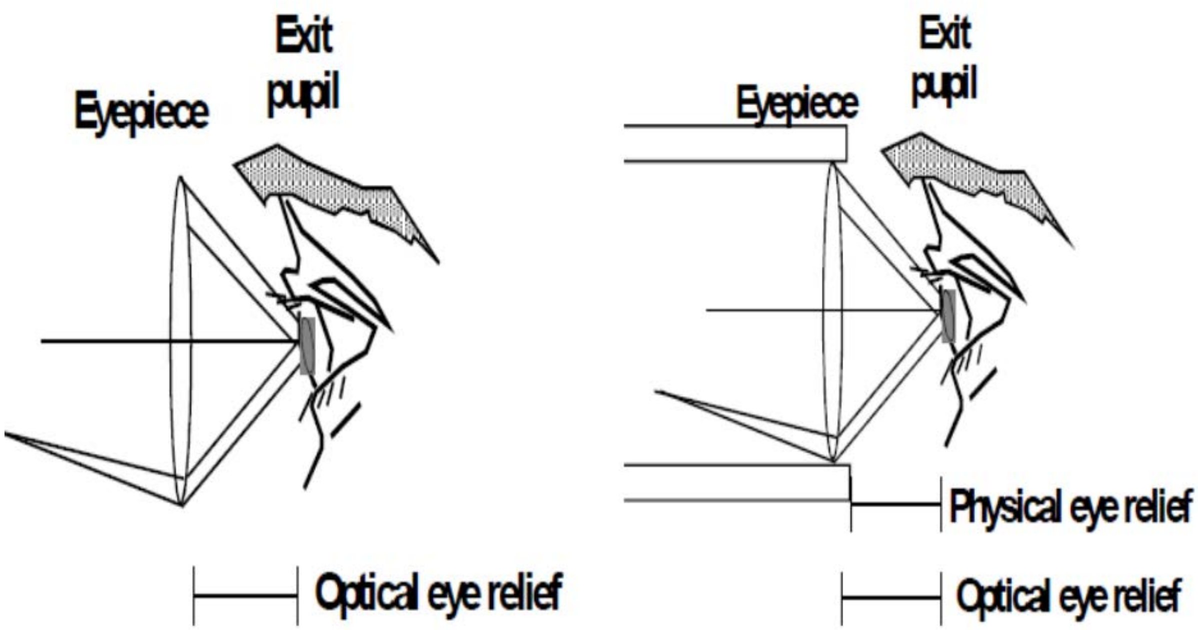

Figure 2.3 Optical eye relief (left).

- Field-of-view (FOV)
- Exit pupil (eye box) size and shape
- Optical eye relief (is defined as the distance from the last optical element to the exit pupil)
- Physical eye relief
- Transmission (optical throughput)

- Beam splitter transmission/reflection coefficients (for see-through HMDs)
- Modulation transfer function (MTF)
- Chromatic aberration
- Distortion
- Field curvature
- Magnification
- Ghosting
- Weight (Mass)
- Center-of-mass (CM)
- Volume (Space required)

2.4 STUDY ON OLED DISPLAY

2.4.1 Introduction

OLED is a solid-state device composed of thin films of organic molecules that create light with the application of electricity. OLEDs can provide brighter, crisper displays on electronic devices and use less power than conventional light emitting diodes(LEDs)used today.

Like an LED, an OLED is a solid-state semiconductor device that is 100 to 500 nanometers thick or about 200 times smaller than a human hair. OLEDs can have either two layers or three layers of organic material. An OLED consist of the following parts:

- Substrate
- Anode
- Organic layers
- Cathode

OLEDS emit light through a process called electro-phosphorescene. Different types of OLEDs are

- Passive-matrix OLED
- Active-matrix OLED
- Transparent OLED

- Foldable OLED
- Top-emitting OLED

Currently, OLEDs are used in small-screen devices such as cell phones, PDAs and digital cameras. Research and development in the field of OLEDs is proceeding rapidly now and may lead to future applications in heads-up displays, automotive dash boards etc.

2.4.1.1 Features of OLED

Organic LED has several inherent properties that afford unique possibilities

- High brightness is achieved at low drive voltages/current densities
- Operating lifetime exceeding 10,000 hours
- Materials do not need to be crystalline, so easy to fabricate
- Possible to fabricate on glass and flexible substrates
- Self luminescent so no requirement of backlighting
- Higher brightness
- Low operating and turn-on voltage

Low cost of materials and substrates of OLEDs can provide desirable advantages over today's liquid crystal displays(LCDs),

- High contrast
- Low power consumption
- Wide operating temperature range
- Long operating lifetime
- A flexible, thin and lightweight
- Increased brightness
- Faster response time for full motion video   Cost effective manufacturability

2.4.2 History

The first observations of electroluminescence in organic materials were in the early 1950s by A. Bernanose and co-workers at the Nancy-Université, France. They applied high-voltage alternating current (AC) fields in air to materials such as acridine orange, either deposited on or dissolved in cellulose or cellophane thin films. The proposed mechanism was either direct excitation of the dye molecules or excitation of electrons. In 1960, Martin Pope and co-workers at New York University developed ohmic dark-injecting electrode contacts to organic crystals. They further described the necessary energetic requirements (work functions) for hole and electron injecting electrode contacts. These contacts are the basis of charge injection in all modern OLED devices. Pope's group also first observed direct current (DC) electroluminescence under vacuum on a pure single crystal of anthracene and on anthracene crystals doped with tetracene in 1963 using a small area silver electrode at 400V. The proposed mechanism was field-accelerated electron excitation of molecular fluorescence. Pope's group reported in 1965 that in the absence of an external electric field, the electroluminescence in anthracene crystals is caused by the recombination of a thermalized electron and hole, and that the conducting level of anthracene is higher in energy than the exciton energy level. Also in 1965, W. Helfrich and W. G. Schneider of the National Research Council in Canada produced double injection recombination electroluminescence for the first time in an anthracene single crystal using hole and electron injecting electrodes, the forerunner of modern double injection devices. In the same year, Dow Chemical researchers patented a method of preparing electroluminescent cells using high voltage (500–1500 V) AC-driven (100–3000 Hz) electrically-insulated one millimeter thin layers of a melted phosphor consisting of ground anthracene powder, tetracene, and graphite powder. Their proposed mechanism involved electronic excitation at the contacts between the graphite particles and the anthracene molecules. Device performance was limited by the poor electrical conductivity of contemporary organic materials. This was overcome by the discovery and development of highly conductive polymers. For more on the history of such materials, see conductive polymers. Electroluminescence from polymer films was first observed by Roger Partridge at the National Physical Laboratory in the United Kingdom. The device consisted of a film of poly(n-vinylcarbazole) up to 2.2 micrometres thick located between two charge injecting electrodes. The results of the project were patented in 1975[14 and published in 1983. The first diode device was reported at Eastman Kodak by Ching W. Tang and Steven Van Slyke in 1987.This device used a novel two-layer structure with separate hole transporting and electron transporting layers such that recombination and light emission occurred in the middle of the organic layer. This resulted in a reduction in operating voltage and improvements in efficiency and led to the current era of OLED research and device production.Research into polymer electroluminescence culminated in 1990 with J. H. Burroughes et al. at the Cavendish Laboratory in Cambridge reporting a high efficiency green light-emitting polymer based device using 100 nm thick films of poly(p-phenylene vinylene).

2.4.3 OLED Composite Layer

Like an LED, an OLED is a solid-state semiconductor device that is 100 to 500 nanometers thick or about 200 times smaller than a human hair, of its deposited film. OLEDs can have either two layers or three layers of organic material; in the latter design, the third layer helps transport electrons from the cathode to the emissive layer. In this article, we will be focusing on the two-organic layer design.

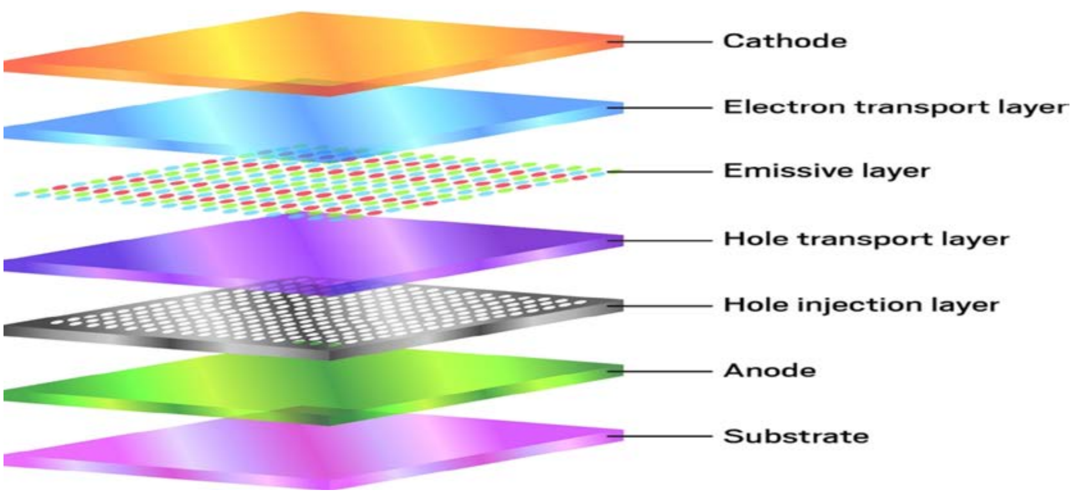

Figure 2.4 General OLED composite layer arrangement

An OLED consists of the following parts:

- Substrate (clear plastic, glass, foil)-The substrate supports the OLED.
- Anode(transparent)-The anode removes electrons (adds electron holes) when a current flow through the device.
- Organic layers- These layers are made up of organic plastic molecules that transport "holes" from the anode. One conducting polymer used in OLEDs is polyaniline.
- Emissive layer- This layer is made of organic plastic molecules (different ones from the conducting layer) that transport electrons from the cathode; this is where light is made. One polymer used in the emissive layer is polyfluorene.

- Cathode (may or may not be transparent depending on the type of OLED)-The cathode injects electrons when a current flow through the device.

2.4.4 OLED OPERATION

OLEDs emit light in a similar manner to LEDs. through a process called electro-phosphorescence

The process is as follows:

- The battery or power supply of the device containing the OLED applies a voltage across the OLED.
- An electrical current flow from the cathode to the anode through the organic layers (an electrical current is a flow of electrons)
    - The cathode gives electrons to the emissive layer of organic molecules.
    - The anode removes electrons from the conductive layer of organic molecules. (This is the equivalent to giving electron holes to the conductive layer)
- At the boundary between the emissive and the conductive layers, electrons find electron holes.
    - When an electron finds an electron hole, the electron fills the hole (it falls into an energy level of the atom that is missing an electron).
    - When this happens, the electron gives up energy in the form of a photon of light.
- The OLED emits light. The color of the light depends on the type of organic molecule in the emissive layer. Manufacturers place several types of organic films on the same OLED to make color displays.
- The intensity or brightness of the light depends on the amount of electrical current applied. The more the current, the brighter the light.

2.4.5 TRANSPARENT OLED

OLEDs are multi-layered light emissive devices that operate of a low applied voltage, typically 3-12Vdc. In an OLEDs simplest form, it is three layer and their most complex, there are five, or more layers. The OLED in this kit is a four-layer device: anode, hole transport layer (HTL), Red Diamond™ emissive layer, cathode. The layers are sandwiched between a plastic substrate and an encapsulation layer. Oxygen and water vapor will slowly migrate through the substrate, degrading the devices performance. The anode is indium tin oxide (ITO) or Aluminum-doped zinc oxide films, that has been pre-deposited onto a PET plastic substrate. The HTL and emissive layers are spin-coated onto the ITO. By altering the spin-coating speed, the thickness of the layers can be altered from 100-500nm. On top of the emissive layer, the cathode is deposited. There are few choices available for the cathode, since it must be a material with a low work-function

2.4.6 Working Principle of OLED

- Substrate (clear plastic, glass, foil)
  - Supports OLED
- Anode
  - Removes electrons
- Conducting layer (P-type material)
  - Organic plastic molecules that transport "holes" from the anode
  - Ex-polyaniline
- Emissive layer (N-type material)
  - Organic plastic molecules (different than the conducting layer) transport electrons from the cathode
  - Light is generated here
  - Ex-polyfluorene
- Cathode
  - Injects electrons

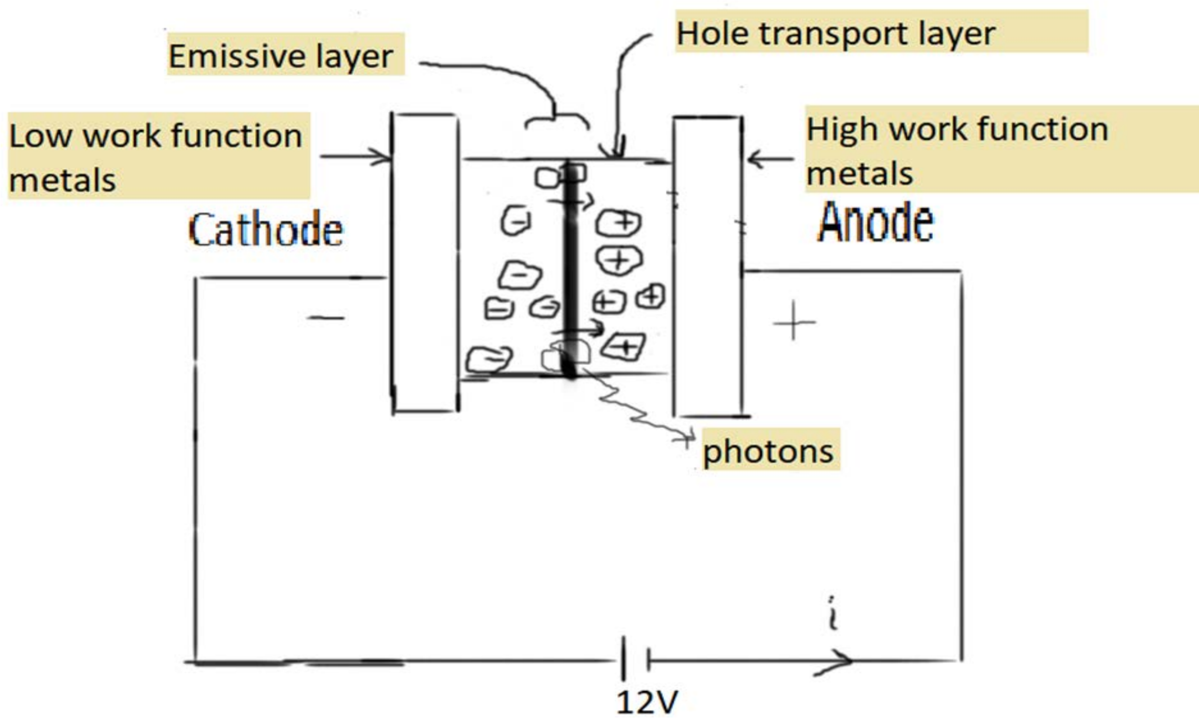

Figure 2.5 working of transparent OLED

During operation, a voltage (Operating voltage: 2-10V) is applied across the OLED such that the anode is positive with respect to the cathode. A current of electrons flows through the device from cathode to anode, as electrons are injected into the LUMO of the organic layer at the cathode and withdrawn from the HOMO at the anode. This latter process may also be described as the injection of electron holes into the HOMO. Electrostatic forces bring the electrons and the holes towards each other and they recombine forming an exciton, a bound state of the electron and hole. This happens closer to the emissive layer, because in organic semiconductors holes are generally more mobile than electrons. The decay of this excited state results in a relaxation of the energy levels of the electron, accompanied by emission of radiation whose frequency is in the visible region. The frequency of this radiation depends on the band gap of the material, in this case the difference in energy between the HOMO and LUMO.

As electrons and holes are fermions with half integer spin, an exciton may either be in a singlet state or a triplet state depending on how the spins of the electron and hole have been combined. Statistically three triplet excitons will be formed for each singlet exciton. Decay from triplet states (phosphorescence) is spin forbidden, increasing the timescale of the transition and limiting the internal efficiency of fluorescent devices. Phosphorescent organic light-emitting diodes make use of spin–orbit interactions to facilitate intersystem crossing between singlet and triplet states, thus obtaining emission from both singlet and triplet states and improving the internal efficiency.

- Trap states
  - Electrons more susceptible to trapping
  - Impurities frequently have empty orbitals (which trap electrons) below -3eV
  - Filled orbitals (which trap holes) above -5eV are not as common

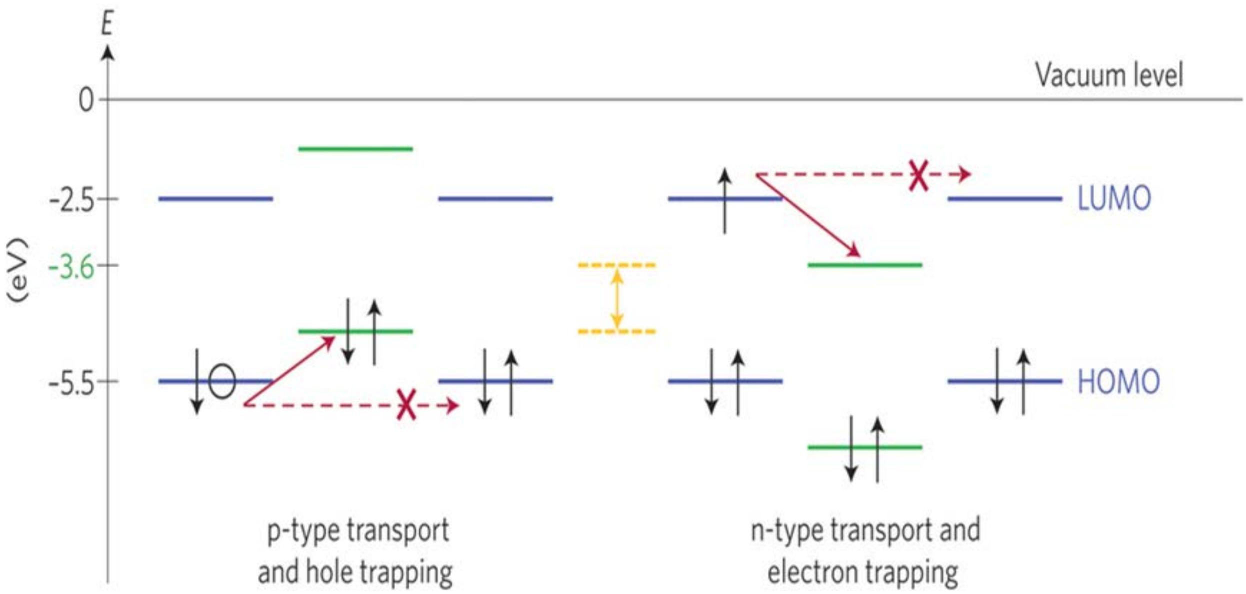

Figure 2.6 Hole and electron trap state

2.5 OUTCOME OF LITERATURE SURVEY

- Design of transparent HMD using OLED is not available in any of the reported literature.
- No efforts have been made to interface the HMD visor with a transparent OLED.
- Though the weight of the OLED deposited film has been assumed to aboub1.5gram. But this can be lighter and thinner, depending on the deposition method used.

3. DESIGN OF DISPLAY ELECTRONIC UNIT

3.1 INTRODUCTION

The alpha numeric interfacing of LM35 with ATmega32 and AT89C51 and 8x8 LED dot matrix display. The interfaced unit has an objective of being able to display digits ranging from (1-50)° Celsius by adjusting the LM35 temperature sensor output, during simulation. The circuit block diagram is shown below:

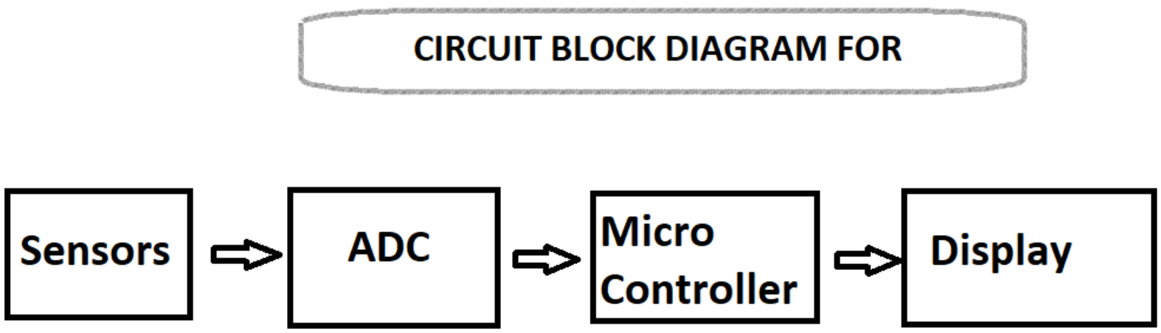

Figure 3.1 Electronic Bock diagram

3.2 Software used

- Proteus
- MCU 8051 IDE
- Atmel Studio 7
- Symbol generator tool (http://gurgleapps.com/tools/matrix)

3.3 Procedure

- Open Proteus software and start a new project
- Select the required components as listed below:
    - One Atmega32 as ADC.
    - One At89c51 as microcontroller.
    - One 8x8 LED matrix for display.
    - One Lm35 as temperature sensor. Every 1°C raise in temperature = 10mV output raise.
    - One 12V DC power supply.

➢ Two 33pF capacitor and One 11.0592MHz crystal oscillator, as clock frequency for the microcontroller.

➢ One push button and one 10KΩ resistor for ADC reset function.

➢ One 10µF(25V) capacitor at AREF pin, so that the external reference voltage (Vref) of the ADC = 5V at AVCC pin. We get a digital output increment for every Vref/2^10 = 5mV.

- Connect (wire) the components as shown below:

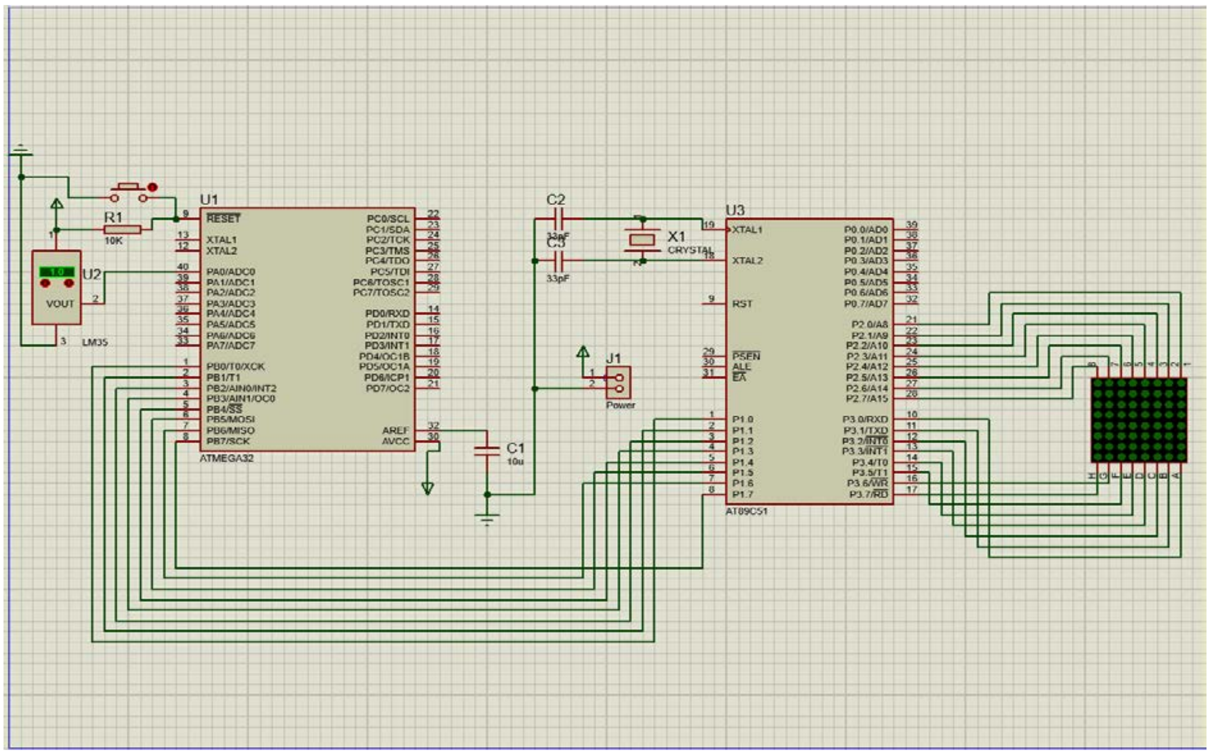

Figure 3.2 Electronics Schematic Diagram

➢ Save project as 'interfaced.pdsprj'

- Open MCU8051IDE software and type the code shown in Appendix 1.

  ➢ Save the work as 'controller.asm' and compile it to 'controller.hex'.

- Open Atmel studio software and type the codec shown in Appendix 2.

  ➢ Save the work as 'adc.c' and compile it to 'adc.hex'.

- Open the saved Proteus project ('interfaced.pdsprj') and import 'adc.hex' file into the Atmage32 and 'controller.hex' into the At89c51, as program files.

- Run the simulation and vary the Lm35 output value, to see the corresponding temperature display in degree Celsius.
- Stop the simulation and save the project again.

3.4 Result

The screenshot of two separate readings are shown below.

- For 35°C

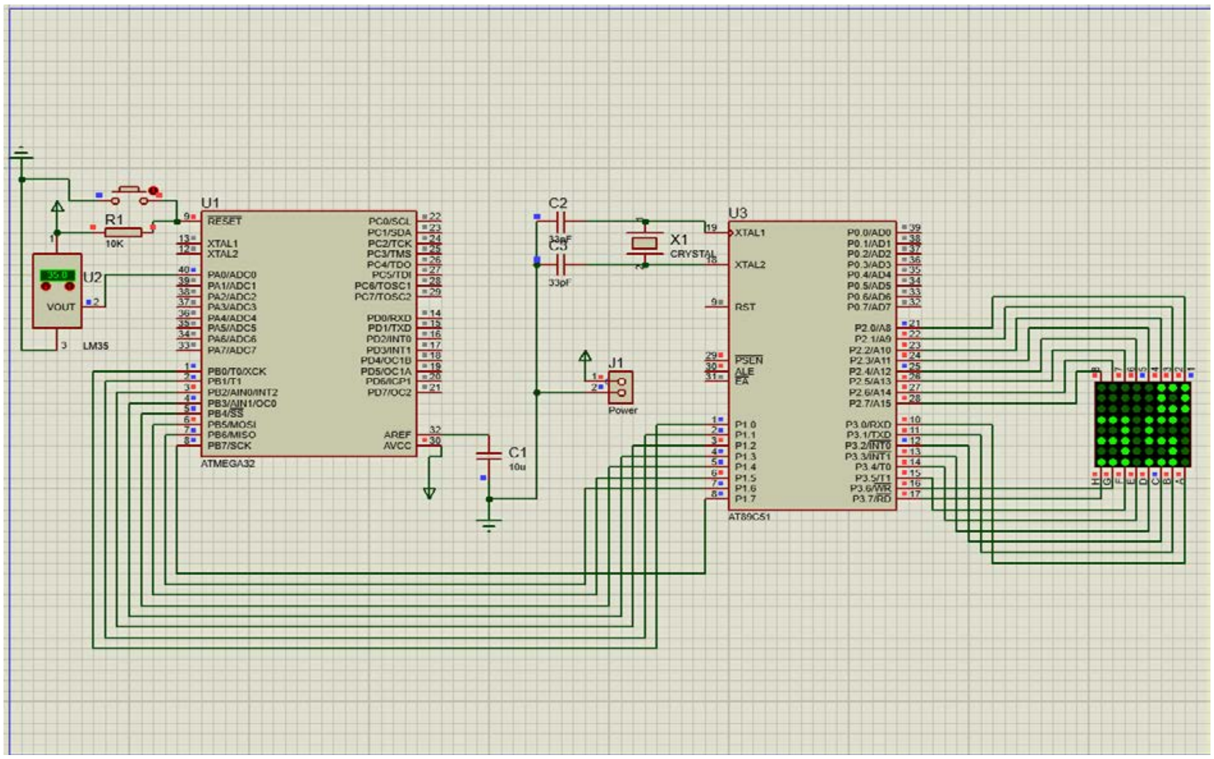

Figure 3.3 First Simulation Result

- For 50°C

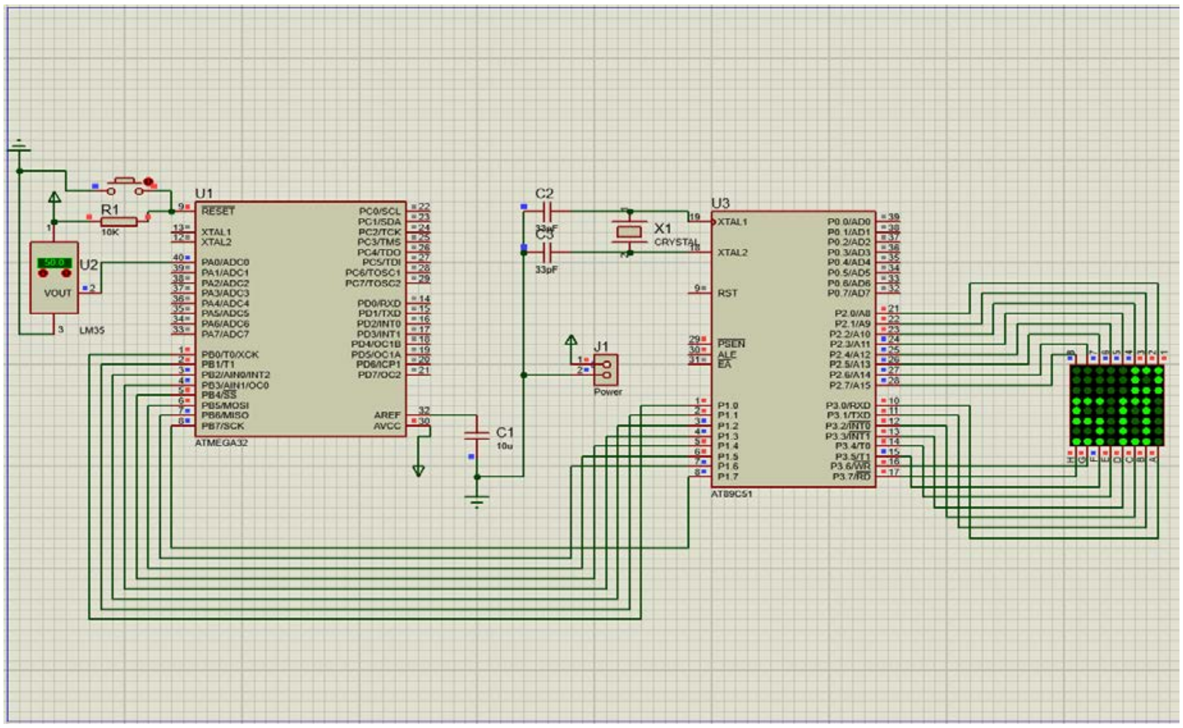

Figure 3.4 Second Simulation Result

3.5 Conclusion

The electronics unit of the trans-missive OLED display is successfully designed and simulated for symbol generation on a traditional 8x8 LED matrix display, which is to be replaced with an 8x8 transparent OLED matrix display.

N/B: Part of the assembly code done, required the bit mapping of the 8x8 matrix display.

4. MATERIAL TECHNOLOGY

4.1 Introduction

The following materials were chosen for the design of transparent OLED. The thickness of the deposited layers (film) can be altered from 100-500nm, by varying the spin-coating speed on the glass substrate. Note that the ITO layer usually comes pre-deposited on the glass substrate. And also, the overall turn on voltage of about 4V, with an overall emissive color of reddish-orange.

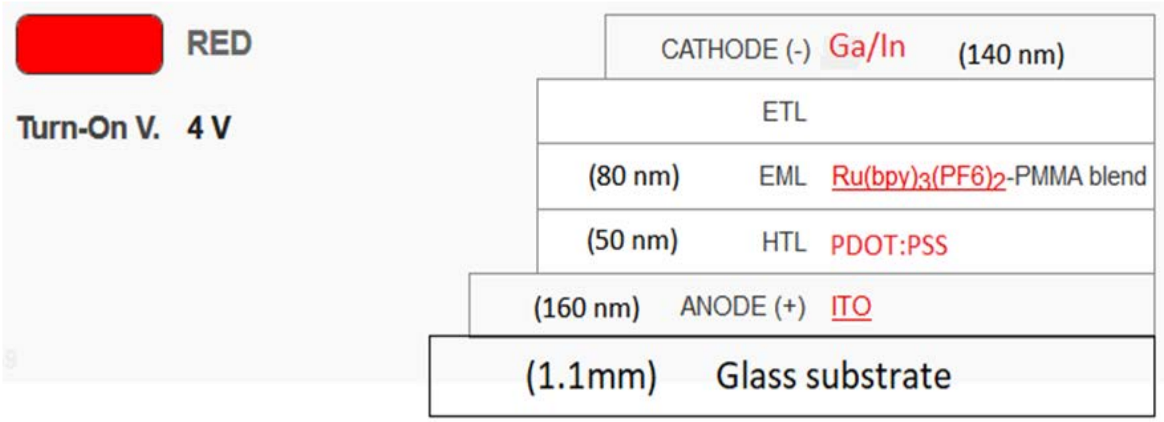

Figure 4.1 Transparent OLED film layout

4.2 Effective Material Selection

The below table shows the cost, size and appearance of the chemicals to be employed for the OLED design.

Table 4.1 List of selected materials and their size, color, form and cost

Material	size	colour	form	Cost (rupees)
1. Poly(ethylenedioxythiophene)/poly(styrenesulfonate) (PEDOT:PSS), as a hole injection layer.	5g	blue	liquid	3,500
2. Tris(2,2′-bipyridine)ruthenium(II) hexafluorophosphate, as an emitting layer.	1g	orange	liquid	12,500

3. Indium-tin-oxide (ITO) for an anode.	20x20 x1.1 mm	colou rless	liqui d	1,500 (20x20x 1.1 mm)
4. Gallium–Indium eutectic alloy, as cathode.	5g	grey	liqui d	7,500 (5g)
Total				25,000

4.2.1 Indium Tin Oxide (ITO)

Is a ternary composition of indium, tin and oxygen in varying proportions. Depending on the oxygen content, it can either be described as a ceramic or alloy. Indium tin oxide is typically encountered as an oxygen-saturated composition with a formulation of 74% In, 18% O2, and 8% Sn by weight. Oxygen-saturated compositions are so typical, that unsaturated compositions are termed oxygen-deficient ITO. It is transparent and colorless in thin layers, while in bulk form it is yellowish to grey. In the infrared region of the spectrum it acts as a metal-like mirror.

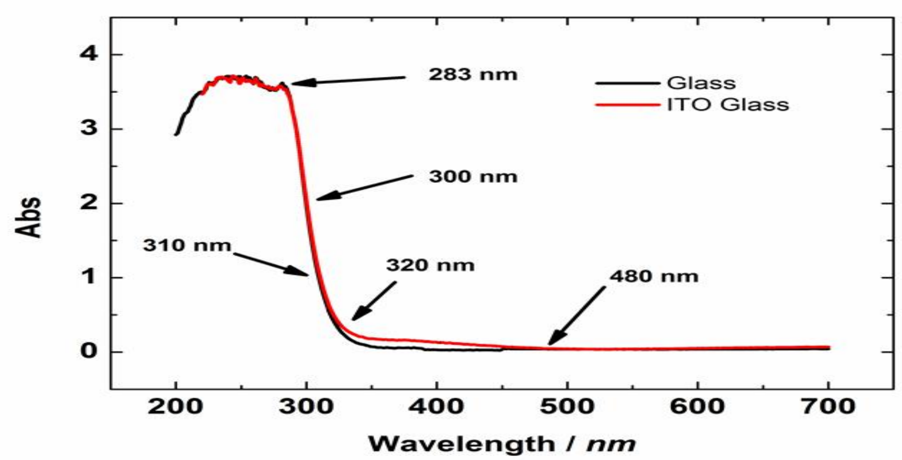

Figure 4.2 Absorption of glass and ITO glass

Indium tin oxide is one of the most widely used transparent conducting oxides because of its two main properties: its electrical conductivity and optical transparency, as well as the ease with which it can be deposited as a thin film. As with all transparent conducting films, a compromise must be made between conductivity and transparency, since increasing the thickness and increasing the concentration of charge carriers increases the material's conductivity, but decreases its transparency.

4.2.2 PDOT:PSS

poly(3,4-ethylenedioxythiophene) polystyrene sulfonate is a polymer mixture of two ionomers. One component in this mixture is made up of sodium polystyrene sulfonate which is a sulfonated polystyrene. Part of the sulfonyl groups are deprotonated and carry a negative charge. The other component poly(3,4-ethylenedioxythiophene) or PEDOT is a conjugated polymer and carries positive charges and is based on polythiophene. Together the charged macromolecules form a macromolecular salt. It is used as a transparent, conductive polymer with high ductility in different applications. For example, AGFA coats 200 million photographic films per year [citation needed] with a thin, extensively-stretched layer of virtually transparent and colorless PEDOT:PSS as an antistatic agent to prevent electrostatic discharges during production and normal film use, independent of humidity conditions, and as electrolyte in polymer electrolytic capacitors. If organic compounds, including high boiling solvents like methylpyrrolidone, dimethyl sulfoxide, sorbitol, ionic liquids and surfactants, are added conductivity increases by many orders of magnitude.

This makes it also suitable as a transparent electrode, for example in touchscreens, organic light-emitting diodes, flexible organic solar cells and electronic paper to replace the traditionally used indium tin oxide (ITO). Owing to the high conductivity (up to 4600 S/cm), it can be used as a cathode material in capacitors replacing manganese dioxide or liquid electrolytes. It is also used in organic electrochemical transistors. The conductivity of PEDOT:PSS can also be significantly improved by a post-treatment with various compounds, such as ethylene glycol, dimethyl sulfoxide (DMSO), salts, zwitterions, co-solvents, acids, alcohols, phenol, geminal diols and amphiphilic fluoro- compounds. This conductivity is comparable to that of ITO, the popular transparent electrode material, and it can triple that of ITO after a network of carbon nanotubes and silver nanowires is embedded into PEDOT:PSS and used for flexible organic devices.

4.2.3 Red Diamond Ruthenium Complex

This an organic synthesized emission compound know as Tris(2,2′-bipyridine)ruthenium(II) hexafluorophosphate, Ru(bpy)3(PF6)2.

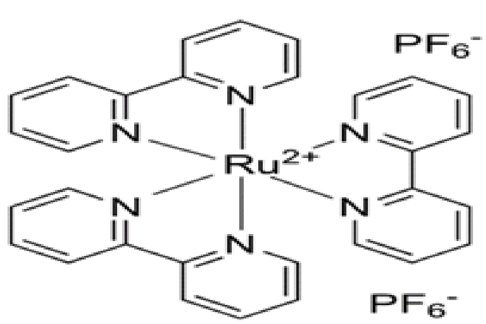

Figure 4.3 Ruthenium Organic structure

This compounds not only exhibit fluorescent mode but also phosphorescent mode, as shown below:

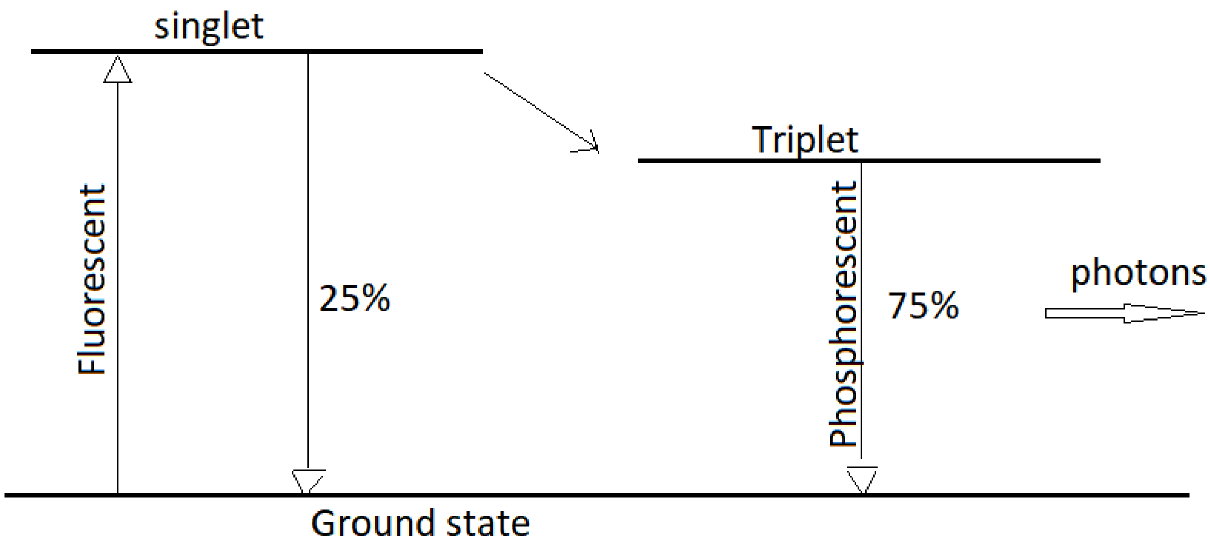

Figure 4.4 Quantum efficiency of phosphorescent mode

Due to this it could reach quantum efficiency of about 100%.

4.2.4 Gallium–Indium Eutectic Alloy

Galinstan is a brand-name and a common name for a liquid metal alloy whose composition is part of a family of eutectic alloys mainly consisting of gallium, indium, and tin. Such eutectic alloys are liquids at room temperature, typically melting at +11 °C (52 °F), while commercial Gallistan melts at −19 °C (−2 °F). An example of a typical eutectic mixture is 68% Ga, 22% In, and 10% Sn (by weight) though proportions vary between 62–95% Ga, 5–22% In, 0–16% Sn (by weight) while remaining eutectic; the exact composition of the commercial product “Galinstan” is not publicly known. Due to the low toxicity and low reactivity of its component metals, galinstan finds use as a replacement for many applications that previously employed the toxic liquid mercury or the reactive NaK (sodium–potassium alloy).

4.2.4.1 Physical Properties

- Boiling point: > 1300°C
- Melting point: −19°C
- Vapour pressure: < 10−8 Torr (at 500°C)
- Density: 6.44 g/cm3 (at 20°C)
- Solubility: Insoluble in water or organic solvents
- Viscosity: 0.0024 Pa·s (at 20°C)
- Thermal conductivity: 16.5 W·m−1·K−1
- Electrical conductivity: 3.46×106 S/m (at 20°C)
- Surface tension: s = 0.535 N/m (at 20°C)

Galinstan tends to be "wet" and adhere to many materials, including glass, which limits its use compared to mercury.

4.3 Charge Transport Mechanisms

Charge transport mechanisms are theoretical models that aim to quantitatively describe the electric current flow through a given medium.

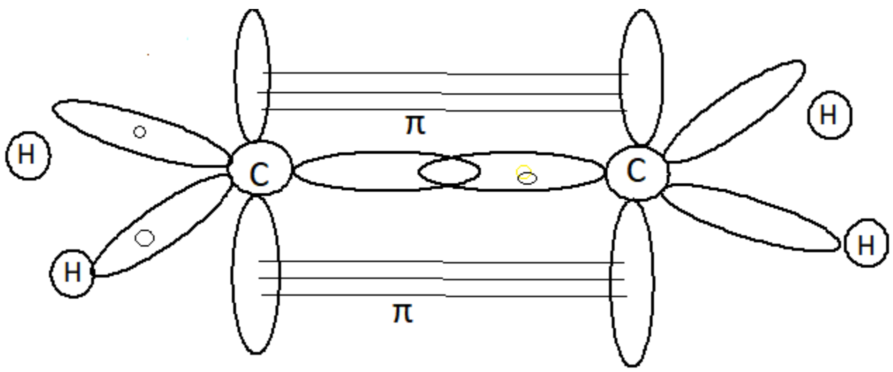

Figure 4.5 Hopping charge transport

Molecular solids exhibit inter-molecular transport, also known as hopping transport.

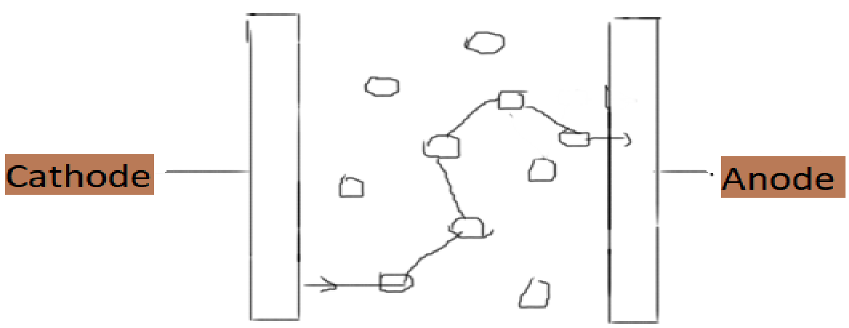

Figure 4.6 Random path of charges

Table 4.2 Characteristics of Hopping Transport

Parameter	Hopping Transport
examples	disordered solids, polycrystalline and amorphous semiconductors

Underlying mechanism	Transition between localized sites via tunnelling (electrons) or overcoming potential barriers (ions)
Inter-site distance	Typically, more than 1 nm
Mean free path	Inter-site distance
mobility	Typically, smaller than 0.01 cm2/Vs; depends on electric field; increases with increasing temperature

4.4 Conclusion

The effective material selection was carried for a semi-transparent OLED display. The characteristic of chosen materials was properly studied for it overall efficiency.

5. CONCEPTUAL MODELLING OF OLED HMD VISOR

5.1 Introduction

The conceptual design of the HMD visor model is carried out using the tools listed below. This design can be reproduced in real time as given in the below detailed steps. First, we need to understand that, there are different deposition method available for better precision and result, such as Vacuum Thermal Evaporation, Organic vapor phase deposition, Inkjet printing method. But for cost effectiveness we are going to be considering the spin-coating method.

5.1.1 Spin Coating

In this method, the liquid organic material is deposited on a glass substrate in excess. Rotating the substrate at a high velocity will cause the liquid to spread out across the required portion as shown in Fig. 5.1 This forms a thin transparent layer of the material and will bake as it evaporates. The thickness of the film is determining factor for the thickness of the film is the amount of time expended in rotating the substrate and drying rate of the organic material in use. Films produced this way usually have a problem of inconsistent thickness, as well as poor a surface finish. Because of this non-uniformity, it is not the method of choice, but the cheapest and simplest method of deposition.

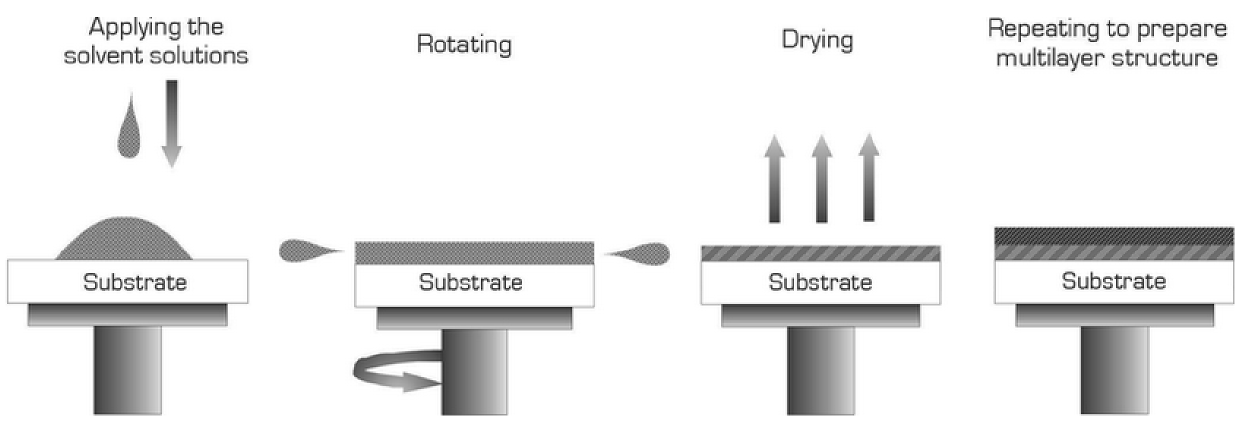

Figure 5.1 The stages involved in the deposition of thin films by spin coating method

5.1.2 Software Used for Model Design

- Solidworks v2017
- Proteus PCB tool v8

5.2 Procedure

- Open Solidworks software and start a new part modelling.

- The ITO glass substrate is first modelled using the following dimensions (20x20x1.1)mm for Length, Breadth and Thickness, respectively. The distance between each column strap is 0.5mm.
    - The extrude boss command is used to create the solid glass part
    - Glass appearance is applied unto this body to give it a transparent texture.
    - Save this part as ‘glass substrate.sldprt’.

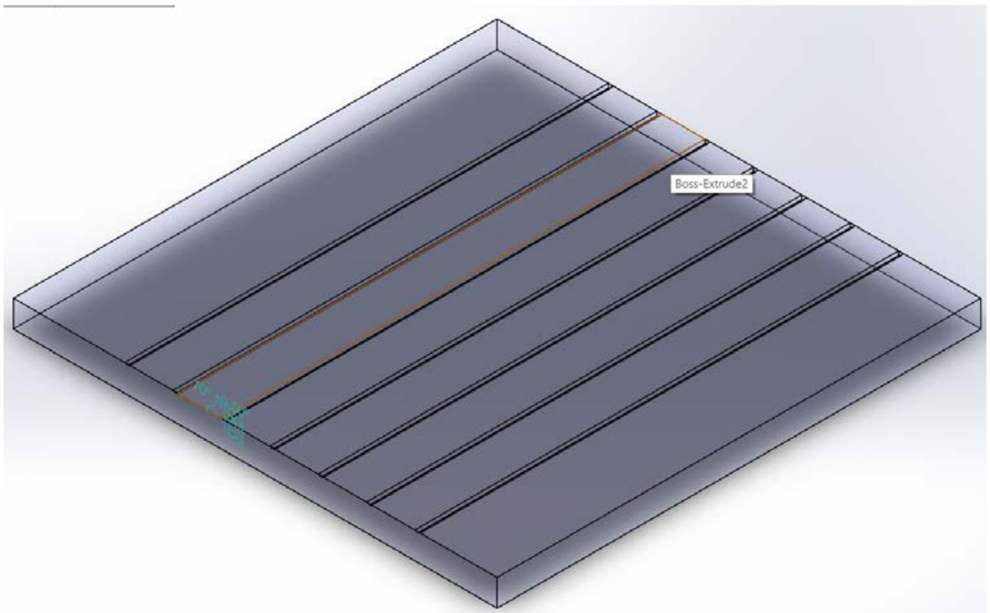

Figure 5.2 Modelled Glass Substrate Part

- Modelling of the deposition layer is done for an 8x8 matrix display. The dimension of each pixel pattern is (1.5x1.5x1.5)mm, and the gap between each pattern pixel is 0.5mm. The overall length and breadth of the pattern layer must be more than (20x20)mm, and the thickness can vary based on the desired film thickness outcome, which can be about 1000nm.

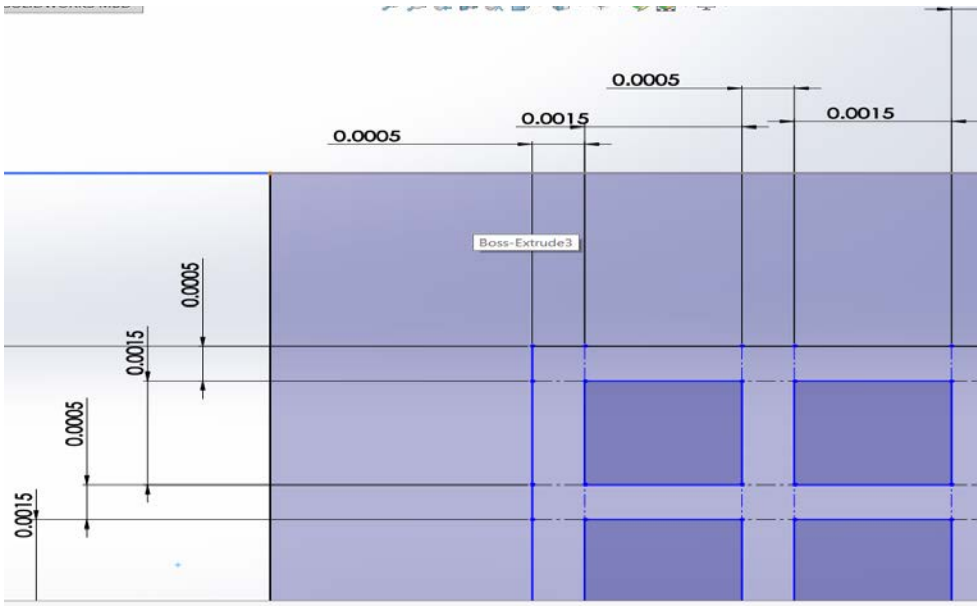

Figure 5.3 Dimensions for Pixel Pattern Model

- After sketching the surface, then extrude boss command is use make the solid patterning surface.
- Extrude cut command is used to create the pixel pattern hole, which is in form of an 8x8 matrix, as shown below.

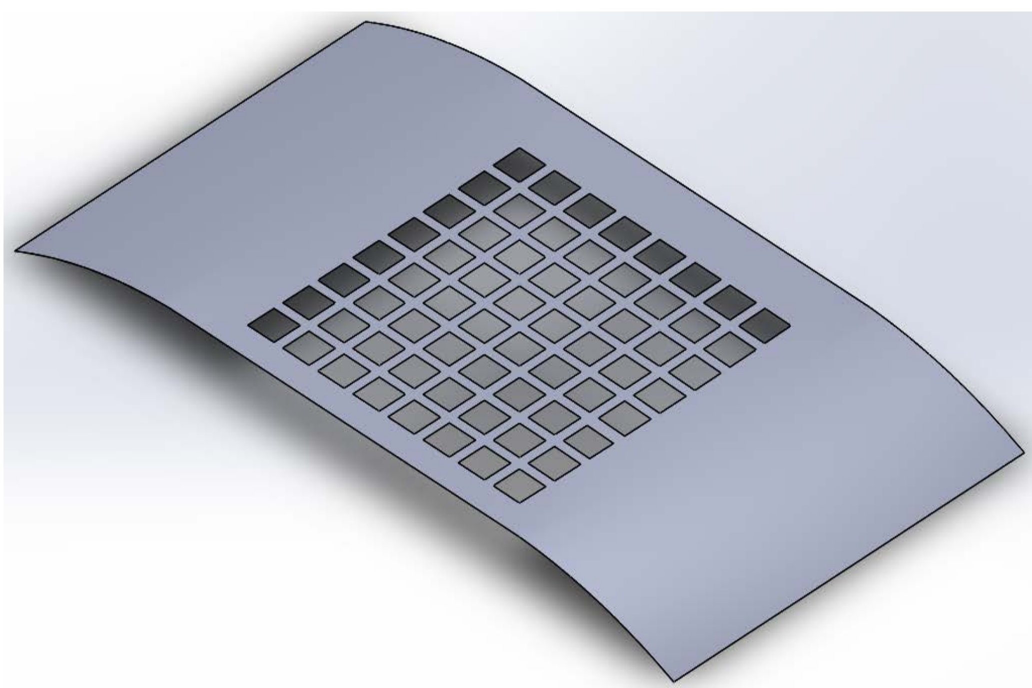

Figure 5.4 Modelled Deposition Patterning Layer Part

- Save this part as 'pattern layer.sldprt'.

- The ‘glass substrate.sldprt’ part and the ‘pattern layer.sldprt’ part are assembled and mated together, by using the Solidworks parts assemble tool.
- Open Proteus PCB tool and generate a 3D M-CAD file with ‘filename’.STEP from the earlier electronics circuit design. Import this file into SolidWorks and assemble it with the transparent OLED model. Mate this parts as shown below.

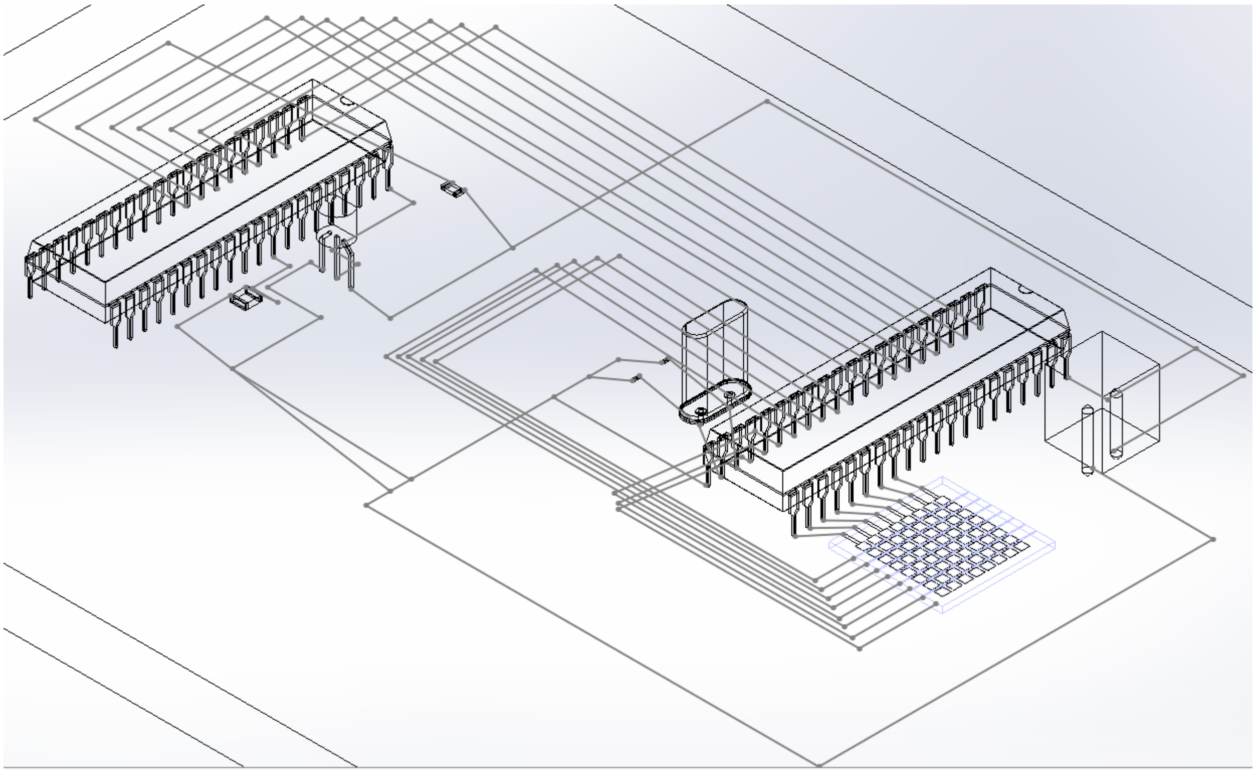

Figure 5.5 Schematic overview of the assembled model

5.3 Procedure for a Conceptual Model

- This should be done using an eye magnifying glass (at least 10X power), for better precision and result. This is due to the size of dot pixel to be created on the glass substrate.
- The glass is to be properly cleaned with ethanol or any other good cleaning agent.
- Carefully strap the glass substrate into eight columns using groove cutting tools of about 70°±0.05° of V-notch, to cut into a depth of about 0.1mm. This will serve as anode of eight isolated strap.
- The pattern deposition layer is to be carefully taped on the glass substrate, as shown below.

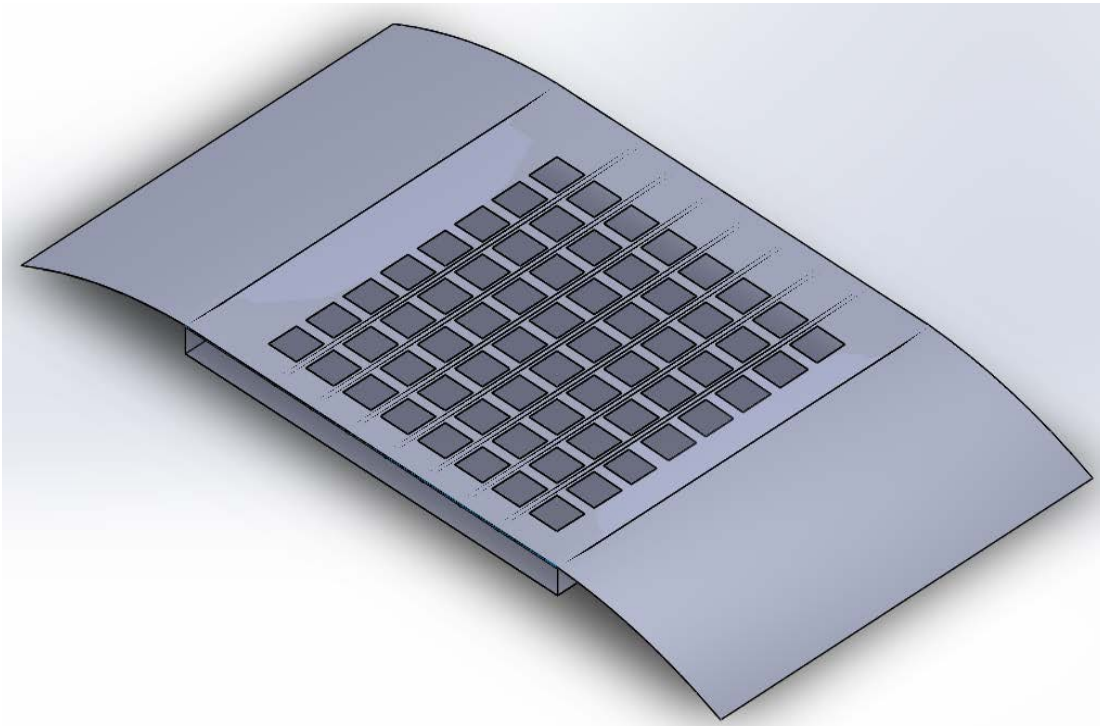

Figure 5.6 Isometric overview of the assembled model

- ➢ The (HIL) PDOT:PSS can be applied using pipettes to each pixels and spin coated (deposited), with the help of any motor at 3000 rpm for 60 s and then the patterning layer (tape) is carefully remove. The result is kept to bake for 30 min at 140°C in an oven. Now the overall appearance is colorless.
- The pattern deposition layer is carefully taped again on the glass substrate.
    - ➢ Then the emissive layer red diamond ruthenium complex is also applied to each pixel and spin-deposited at 3000 rpm for 20 s and then the patterning layer (tape) is carefully removed again. The result is kept to baked at room temperature. Now its overall appearance is orange in color, as shown below.

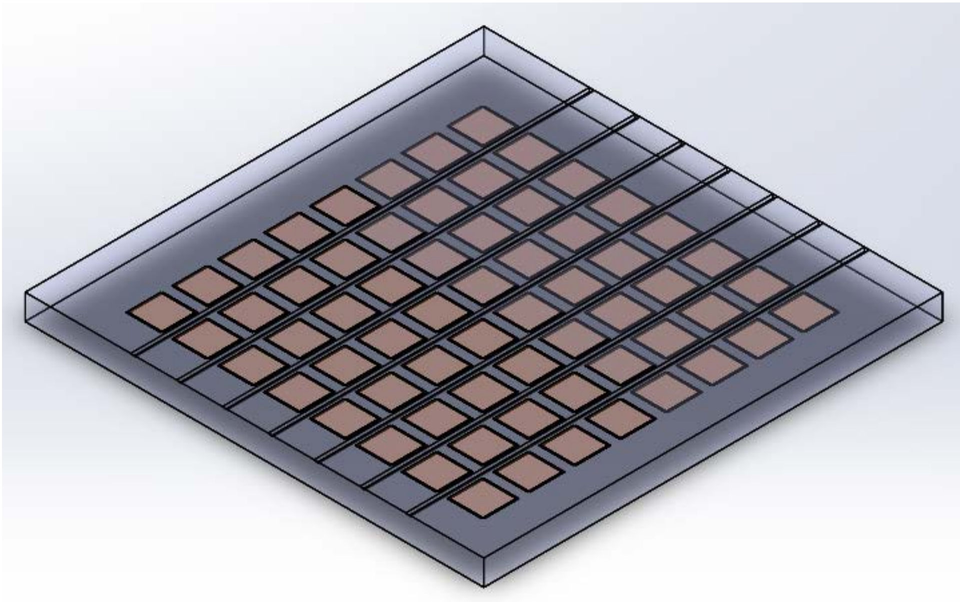

Figure 5.7 Isometric overview of the modelled transparent 8x8 OLED display

- Now, a different pattern of deposition layer is also carefully taped on the glass substrate, as shown below.

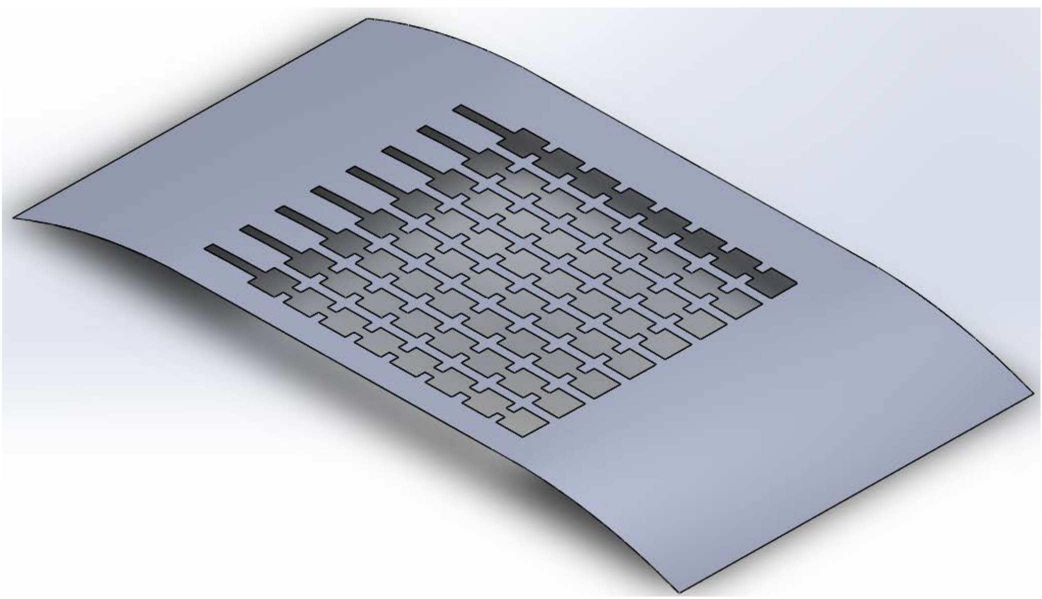

Figure 5.8 Isometric view of the Modelled Cathode deposition layer part

  - The Cathode layer which is the gallium-indium eutectic alloy, is applied to each pixel using this deposition pattern, with a strap of eight rows. This alloy is spin-deposited with speed greater than 3000 rpm for 2 min and allowed to properly dry.

- After making the semi-transparent OLED matrix, the electronics unit is to be interfaced with it through the following steps:
  - ➢ Properly connect the OLED with the electronic unit, through the use of conducting wire on a bread board, as shown in the schematic diagram below.

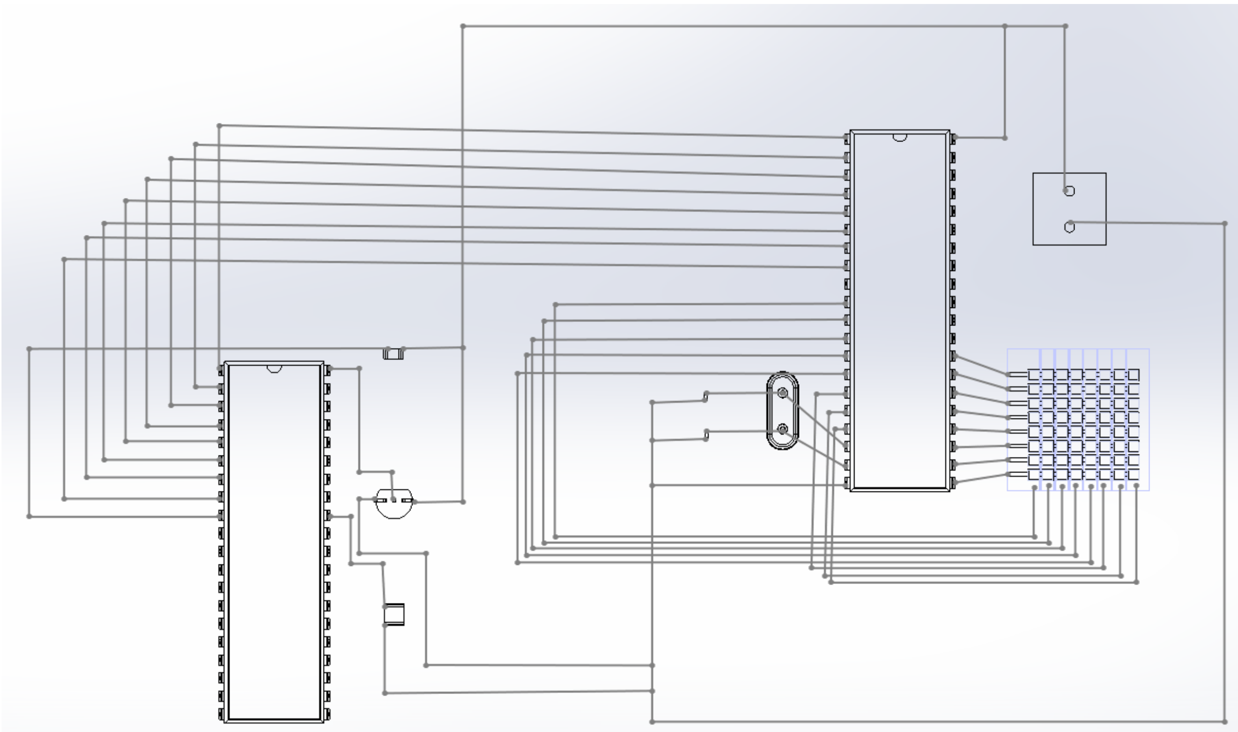

Figure 5.9 Schematic Overview of the Modelled OLED HMD visor

  - ➢ The 8 Anode strips are to be connected from pin 10-17 of the microcontroller.
  - ➢ The 8 Cathode strips are to be connected from pin 21-28 of the microcontroller.
    N/B: The numbering of the microcontroller is from lift to right (anti-clockwise), as shown below:

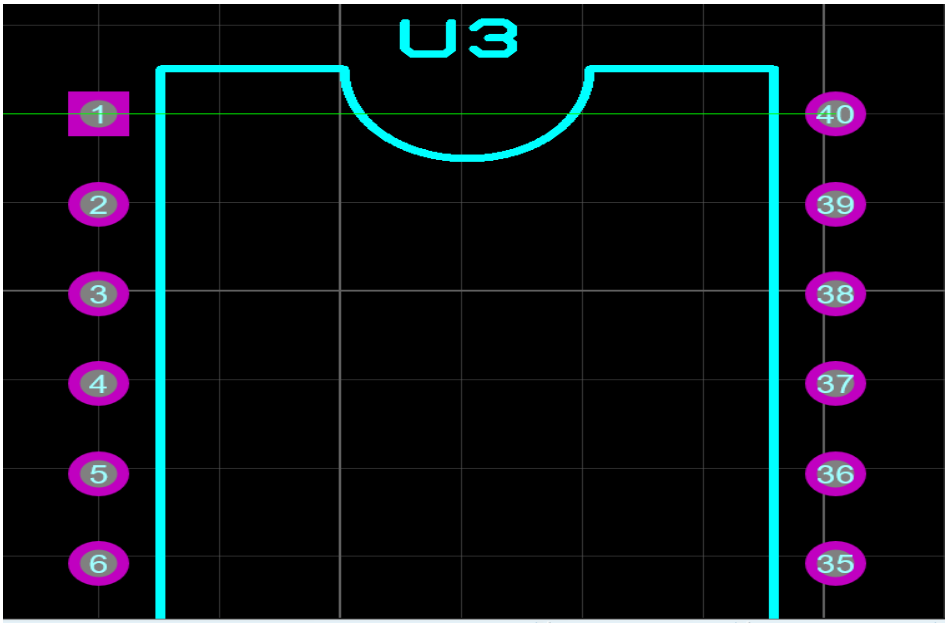

Figure 3.10 Pin arrangement of the Integrated Circuit (IC)

- An AVR USB Programmer(USBASP) is used for loading the assembly program onto the microcontroller through a bootloader software. And connect the 5 output pins from the programmer board to the respective 5 input pins of the microcontroller.
- The list of components used are shown below

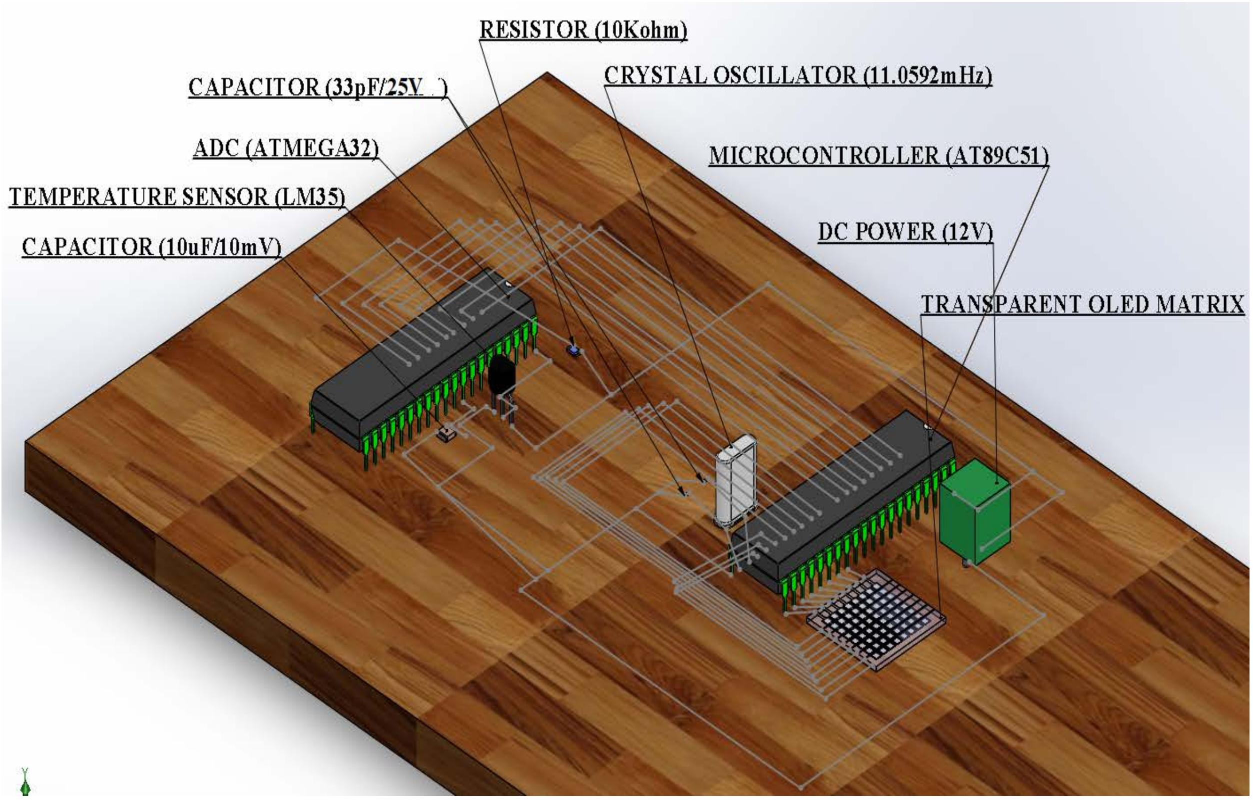

Figure 5.11 Overview of components required for the project

5.4 Conclusion

The conceptual model design was carried with detailed procedures for experimental purpose, of which the expected transmittance would be about (45-70)% for deposited film thickness less than 800 nm.

6. CONCLUSION

The concept for a transparent HMD using an OLED is proven with the following anticipated outcome:

- Light weight overall helmet mounted display
- See plane display of pilot information directly on the visor, which is also a see through device. For observing the external environment
- transmittance of about (45-70)% for a deposited film thickness that is less than 1000 nm.
- Fast generation of display symbols (symbology). Due to the absence of CRT, lens arrangement, partial reflecting mirror visor, etc.
- Therefore, the conceptual proof of a semi-transparent HMD OLED visor was carried out.

SCOPE OF FUTURE WORK

1. Use of more efficient deposition methods link the Nano ink jet printing.
2. Designing electronics circuit to support higher pixel resolution (i.e. higher OLED matrix display), with the help of multiplexers.
3. Fabrication of the Transparent HMD visor, using OLED
4. Synthesizing efficient and cheaper organic materials for better result.
5. Integrating the display with more pilot information, such as pressure, altitude, speed,etc.

REFERENCES

1. Kevin W. Greeley, "Specialized cockpit displays, symbology, and control mechanization for air vehicle weapons aiming and employment", Proc. SPIE 4362, Cockpit Displays VIII: Displays for Defense Applications, (7 September 2001)
2. J.A. Uphaus, R.B. Bryant, “An innovative tool for designing fault tolerant cockpit display symbology”, Cockpit Integration Div., Wright Lab., Wright-Patterson AFB, OH, USA
3. Kaber DB, Alexander AL, Stelzer EM, Kim SH, Kaufmann K, Hsiang S,”Perceived clutter in advanced cockpit displays: measurement and modeling with experienced pilots”
4. J. Sheats, H. Antoniadis, M. Hueschen, et al, “Organic electroluminescent,” Science 273, 884-888 (1996).

5. C. Tang, and S. VanSlyke, "Organic Electroluminescent Diodes," Applied Physics Letters 51, 913-915 (1987).
6. Wikipedia contributors, "Spin Coating," Wikipedia, the free encyclopedia, http://en.wikipedia.org/wiki/Spin_coating (28 November 2006).
7. W. Helfrich and W. G. Schneider, "Recombination radiation in anthracene crystals," Physical Review Letters 14, 229-231 (1965).
8. Sigma Aldrich contributors, Material science | learning center, Technology spotlight libery, "OLED efabricator" https://www.sigmaaldrich.com/materials-science/learning-center/oled-efabricator.html
9. L.J. Rothberg, and A.J. Lovinger, "Status of and prospects for organic electroluminescence," Journal of Materials Research 11, 3174-3187 (1996).
10. N. Savage, "Going Organic," OE Magazine, http://oemagazine.com/FromtheMagazine/nov03/prodtrends.html (28 November 2006).
11. P. E. Burrows, G. Gu, V. Bulovic, Z. Shen, S. R. Forrest, and M. E. Thompson, "Achieving full-color organic light-emitting devices for lightweight, flat-panel displays," IEEE Transactions on electron devices 44 (8), 1188-1203 (1997).
12. Institute of Chemical Sciences and Engineering, "Device Preparation," http://ssg.epfl.ch/dev-prep.html coating (28 November 2006).
13. J. Kido, and Y. Iizumi, "Fabrication of highly efficient organic electroluminescent devices," Applied Physics Letters 73, 2721-2723 (1998).
14. Sugimoto, H. Ochi, S. Fujimura, et al, "Flexible OLED displays using plastic substrates," IEEE Journal of Selected Topics in Quantum Electronics 10, 107-144

APPENDIX 1

The assembly language code written in the MCU8051IDE software is given below:

```
ORG 0H
        MOV TMOD,#01H
        MOV R2,#11111110B
BACK:       MOV A,P1
ACALL DELAY
ACALL DELAY
ACALL DELAY
ACALL DELAY
COND1:  CJNE A,#01H,COND2
 MOV DPTR,#ONE
 JMP MAIN
COND2: CJNE A,#02H,COND3
 MOV DPTR,#TWO
 JMP MAIN
COND3: CJNE A,#03,COND4
 MOV DPTR,#THREE
 JMP MAIN
COND4: CJNE A,#04H,COND5
 MOV DPTR,#FOUR
 JMP MAIN
COND5: CJNE A,#05H,COND6
 MOV DPTR,#FIVE
 JMP MAIN
COND6: CJNE A,#06H,COND7
 MOV DPTR,#SIX
 JMP MAIN
```

```
COND7: CJNE A,#07H,COND8
  MOV DPTR, #SEVEN
  JMP MAIN
COND8: CJNE A,#08H,COND9
  MOV DPTR, #EIGHT
  JMP MAIN
COND9: CJNE A,#09H,COND10
  MOV DPTR,#NINE
  JMP MAIN
COND10: CJNE A,#0AH,COND11
  MOV DPTR,#TEN
  JMP MAIN
COND11: CJNE A,#0BH,COND12
  MOV DPTR, #ELEVEN
  JMP MAIN
COND12: CJNE A,#0CH,COND13
  MOV DPTR, #TWELVE
  JMP MAIN
COND13: CJNE A,#0DH,COND14
  MOV DPTR, #THIRTEEN
  JMP MAIN
COND14: CJNE A,#0EH,COND15
  MOV DPTR, #FOURTEEN
  JMP MAIN
COND15: CJNE A,#0FH,COND16
  MOV DPTR, #FIFTEEN
  JMP MAIN
COND16: CJNE A,#10H,COND17
  MOV DPTR, #SIXTEEN
  JMP MAIN
COND17: CJNE A,#11H,COND18
```

```
  MOV DPTR, #SEVENTEEN
  JMP MAIN
COND18: CJNE A,#12H,COND19
  MOV DPTR, #EIGHTEEN
  JMP MAIN
COND19: CJNE A,#13H,COND20
  MOV DPTR, #NINETEEN
  JMP MAIN
COND20: CJNE A,#14H,COND21
  MOV DPTR, #TWENTY
  JMP MAIN
COND21: CJNE A,#15H,COND22
  MOV DPTR, #TWENTYONE
  JMP MAIN
COND22: CJNE A,#16H,COND23
  MOV DPTR, #TWENTYTWO
  JMP MAIN
COND23: CJNE A,#17H,COND24
  MOV DPTR,#TWENTYTHREE
  JMP MAIN
COND24: CJNE A,#18H,COND25
  MOV DPTR,#TWENTYFOUR
  JMP MAIN
COND25: CJNE A,#1AH,COND26
  MOV DPTR,#TWENTYFIVE
  JMP MAIN
COND26: CJNE A,#1BH,COND27
  MOV DPTR,#TWENTYSIX
  JMP MAIN
COND27: CJNE A,#1CH,COND28
  MOV DPTR,#TWENTYSEVEN
```

```
 JMP MAIN
COND28: CJNE A,#1DH,COND29
 MOV DPTR,#TWENTYEIGHT
 JMP MAIN
COND29: CJNE A,#1EH,COND30
 MOV DPTR,#TWENTYNINE
 JMP MAIN
COND30: CJNE A,#1FH,COND31
 MOV DPTR, #THIRTY
 JMP MAIN
COND31: CJNE A,#20H,COND32
 MOV DPTR,#THIRTYONE
 JMP MAIN
COND32: CJNE A,#21H,COND33
 MOV DPTR,#THIRTYTWO
 LJMP MAIN
COND33: CJNE A,#22H,COND34
 MOV DPTR,#THIRTYTHREE
 LJMP MAIN
COND34: CJNE A,#23H,COND35
 MOV DPTR,#THIRTYFOUR
 LJMP MAIN
COND35: CJNE A,#24H,COND36
 MOV DPTR,#THIRTYFIVE
 LJMP MAIN
COND36: CJNE A,#25H,COND37
 MOV DPTR,#THIRTYSIX
 SJMP MAIN
COND37: CJNE A,#26H,COND38
 MOV DPTR,#THIRTYSEVEN
 SJMP MAIN
```

```
COND38: CJNE A,#27H,COND39
  MOV DPTR,#THIRTYEIGHTH
  SJMP MAIN
COND39: CJNE A,#28H,COND40
  MOV DPTR,#THIRTYNINE
  SJMP MAIN
COND40: CJNE A,#29H,COND41
  MOV DPTR,#FORTY
  SJMP MAIN
COND41: CJNE A,#2AH,COND42
  MOV DPTR,#FORTYONE
  SJMP MAIN
COND42: CJNE A,#2BH,COND43
  MOV DPTR,#FORTYTWO
  SJMP MAIN
COND43: CJNE A,#2CH,COND44
  MOV DPTR,#FORTYTHREE
  SJMP MAIN
COND44: CJNE A,#2DH,COND45
  MOV DPTR,#FORTYFOUR
  SJMP MAIN
COND45: CJNE A,#2EH,COND46
  MOV DPTR,#FORTYFIVE
  SJMP MAIN
COND46: CJNE A,#2FH,COND47
  MOV DPTR,#FORTYSIX
  SJMP MAIN
COND47: CJNE A,#30H,COND48
  MOV DPTR,#FORTYSEVEN
  SJMP MAIN
COND48: CJNE A,#31H,COND49
```

```
  MOV DPTR,#FORTYEIGHT
  SJMP MAIN
COND49: CJNE A,#32H,COND50
  MOV DPTR,#FORTYNINE
  SJMP MAIN
COND50: CJNE A,#33H,COND0
  MOV DPTR,#FIFTY
  SJMP MAIN
COND0: CJNE A,#0H,BACK1
  MOV DPTR,#ZERO
MAIN:   MOV R3,#050H
LOOP1:        MOV R4,#08H
LOOP:   MOV A,R2
        MOV P3,A
        RL A
        MOV R2,A
        MOV A,R4
        MOVC A,@A+DPTR
        MOV P2,A
        ACALL DELAY
        DJNZ R4,LOOP
        DJNZ R3,LOOP1
        LJMP BACK
DELAY:        MOV TH0,#0FFH
        MOV TL0,#0
        SETB TR0
HERE:   JNB TF0,HERE
        CLR TR0
        CLR TF0
        RET
BACK1:  LJMP BACK
```

```
ORG 0FFFH
ONE:DB
00000000B,00000111B,00000101B,00000111B,01000000B,11000000B,1000000B,0100
0000B,11100000B
TWO:DB 0x0,0x07,0x05,0x07,0xe0,0x20,0xe0,0x80,0xe0
THREE:DB 0x0,0x7,0x5,0x7,0xe0,0x20,0xe0,0x20,0xe0
FOUR:DB 0x0,0x07,0x05,0x07,0xa0,0xa0,0xe0,0x20,0x20
FIVE:DB 0x0,0x07,0x05,0x07,0xe0,0x80,0xe0,0x20,0xe0
SIX:DB 0x0,0x07,0x05,0x07,0xe0,0x80,0xe0,0xa0,0xe0
SEVEN:DB 0x0,0x07,0x05,0x07,0xe0,0x20,0x20,0x20,0x20
EIGHT:DB 0x0,0x07,0x05,0x07,0xe0,0xa0,0xe0,0xa0,0xe0
NINE:DB 0x0,0x07,0x05,0x07,0xe0,0xa0,0xe0,0x20,0xe0
TEN:DB 0x0,0x07,0x05,0x07,0x5c,0xd4,0x54,0x54,0x5c
ELEVEN:DB 0x0,0x07,0x05,0x07,0x48,0xd8,0x48,0x48,0x48
TWELVE:DB 0x0,0x07,0x05,0x07,0x5c,0xc4,0x5c,0x50,0x5c
THIRTEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xc4,0x5c,0x44,0x5c
FOURTEEN:DB 0x0,0x07,0x05,0x07,0x54,0xd4,0x5c,0x44,0x44
FIFTEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xd0,0x5c,0x44,0x5c
SIXTEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xd0,0x5c,0x54,0x5c
SEVENTEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xc4,0x44,0x44,0x44
EIGHTEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xd4,0x5c,0x54,0x5c
NINETEEN:DB 0x0,0x07,0x05,0x07,0x5c,0xd4,0x5c,0x44,0x5c
TWENTY:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xea,0x8a,0xee
TWENTYONE:DB 0x0,0x07,0x05,0x07,0xe4,0x2c,0xe4,0x84,0xe4
TWENTYTWO:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xee,0x88,0xee
TWENTYTHREE:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xee,0x82,0xee
TWENTYFOUR:DB 0x0,0x07,0x05,0x07,0xea,0x2a,0xee,0x82,0xe2
TWENTYFIVE:DB 0x0,0x07,0x05,0x07,0xee,0x28,0xee,0x82,0xee
TWENTYSIX:DB 0x0,0x07,0x05,0x07,0xee,0x28,0xee,0x8a,0xee
TWENTYSEVEN:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xe2,0x82,0xe2
TWENTYEIGHT:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xee,0x8a,0xee
```

```
TWENTYNINE:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xee,0x82,0xee
THIRTY:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xea,0x2a,0xee
THIRTYONE:DB 0x0,0x07,0x05,0x07,0xe4,0x2c,0xe4,0x24,0xe4
THIRTYTWO:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xee,0x28,0xee
THIRTYTHREE:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xee,0x22,0xee
THIRTYFOUR:DB 0x0,0x07,0x05,0x07,0xea,0x2a,0xee,0x22,0xe2
THIRTYFIVE:DB 0x0,0x07,0x05,0x07,0xee,0x28,0xee,0x22,0xee
THIRTYSIX:DB 0x0,0x07,0x05,0x07,0xee,0x28,0xee,0x2a,0xee
THIRTYSEVEN:DB 0x0,0x07,0x05,0x07,0xee,0x22,0xe2,0x22,0xe2
THIRTYEIGHTH:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xee,0x2a,0xee
THIRTYNINE:DB 0x0,0x07,0x05,0x07,0xee,0x2a,0xee,0x22,0xee
FORTY:DB 0x0,0x07,0x05,0x07,0xae,0xaa,0xea,0x2a,0x2e
FORTYONE:DB 0x0,0x07,0x05,0x07,0xa4,0xac,0xe4,0x24,0x24
FORTYTWO:DB 0x0,0x07,0x05,0x07,0xae,0xa2,0xee,0x28,0x2e
FORTYTHREE:DB 0x0,0x07,0x05,0x07,0xae,0xa2,0xee,0x22,0x2e
FORTYFOUR:DB 0x0,0x07,0x05,0x07,0xaa,0xaa,0xee,0x22,0x22
FORTYFIVE:DB 0x0,0x07,0x05,0x07,0xae,0xa8,0xee,0x22,0x2e
FORTYSIX:DB 0x0,0x07,0x05,0x07,0xae,0xa8,0xee,0x2a,0x2e
FORTYSEVEN:DB 0x0,0x07,0x05,0x07,0xae,0xa2,0xe2,0x22,0x22
FORTYEIGHT:DB 0x0,0x07,0x05,0x07,0xae,0xaa,0xee,0x2a,0x2e
FORTYNINE:DB 0x0,0x07,0x05,0x07,0xae,0xaa,0xee,0x22,0x2e
FIFTY:DB 0x0,0x07,0x05,0x07,0xee,0x8a,0xea,0x2a,0xee
ZERO:DB 0x0,0x07,0x05,0x07,0x6e,0x8a,0x6a,0x0a,0xee
END
```

APPENDIX 2

The c language code written in the Atmel Studio7 is given below:

```
/*
 * filname          : adc.c
 * Created          : 07/04/2018 02:51:32 PM
 * Author           : Ezeorah Godswill
 *Microcontroller   : ATmega32
 *System Clock              : 1MHz
 *Project                   : LM35 Temperature Sensor Interfacing with ATmega32 and
LED 8x8 matrix Display
*/
#include<avr/io.h>
/*Includes io.h header file where all the Input/Output Registers and its Bits are defined
for all AVR microcontrollers*/

#defineF_CPU            1000000
/*Defines a macro for the delay.h header file. F_CPU is the microcontroller frequency
value for the delay.h header file. Default value of F_CPU in delay.h header file is
1000000(1MHz)*/

#include<util/delay.h>
/*Includes delay.h header file which defines two functions, _delay_ms (millisecond
delay) and _delay_us (microsecond delay)*/

/*ADC Function Declarations*/
void adc_init(void);
int read_adc_channel(unsigned char channel);
```

```
int main(void)
{
	DDRB=0xff;
	/*All the 8 pins of PortB are declared output (LED array is connected)*/

	int adc_output,temperature;
	/*Variable declarations*/

	adc_init();
	/*ADC initialization*/

	/*Start of infinite loop*/
	while(1)
	{
		adc_output=read_adc_channel(0);
		/*Reading temperature sensor value*/

		temperature=adc_output/2;
		/*Temperature value in degree centigrade*/

		PORTB=temperature;
		/*Upper 2 bit of temperature is displayed in LED array*/
	}
}
/*End of Program*/

/*ADC Function Definitions*/
void adc_init(void)
{
	ADCSRA=(1<<ADEN)|(1<<ADSC)|(1<<ADATE)|(1<<ADPS2);
	SFIOR=0x00;
```

```
}

int read_adc_channel(unsigned char channel)
{
        int adc_value;
        unsigned char temp;
        ADMUX=(1<<REFS0)|channel;
        _delay_ms(1);
        temp=ADCL;
        adc_value=ADCH;
        adc_value=(adc_value<<8)|temp;
        return adc_value;
}
```

Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES

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Karunya Nagar, Coimbatore 641 114, Tamil Nadu, India.

Department of Aerospace Engineering

Declaration

Academic Year:	Title of the Project:	Name of the Guide:
2017-2018	PROOF OF CONCEPT FOR TRANSPARENT HMD USING OLED	Lt. Musica SR.

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Reg. No.: **UR14AE067**

Signature of the candidate:

PAPER PRESENTATION & PUBLICATION

1. Ezeorah Godswill, Lt. Musica SR., “Design of HMD symbology using OLED”, 8th International Conference on Science and Engineering (ICSIE 2018) on 1st April 2018.

CONFERENCE CERTIFICATE OBTAINED

8TH INTERNATIONAL CONFERENCE ON SCIENCE AND INNOVATIVE ENGINEERING 2018

ORGANIZED BY

ORGANIZATION OF SCIENCE & INNOVATIVE ENGINEERING AND TECHNOLOGY, CHENNAI

IN ASSOCIATION WITH

JAWAHAR ENGINEERING COLLEGE, CHENNAI

Certificate of Presentation

This is to certify that Dr./Mr./Ms. EZEORAH GODS WILL from KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES has presented a paper titled DESIGN OF HMD SYMBOLOGY USING OLED in the "8th International Conference on Science and Innovative Engineering" held on 1st April 2018.

Secretary

Director

PAPER PRESENTED

Design of HMD symbology using OLED

Ezeorah Godswill, Lt. Musica SR., musica@karunya.edu

Abstract:

Head mounted displays are currently used only in military aircrafts with many recent designs been equipped with the capability of projecting image directly on a trans-missive visor with the help of CRT(for image generation).This system has high weight(about 3kg), monocular rivalry (due to image overlap) and high power consumption (due to the use of CRT and its analogue electronics components).The interfacing of an OLED display with a digital electronic circuit will be able to generate symbols at an effective optical eye relief distance (of about 70-100mm) depending on the eye pupil exit. The integration design of this system with HMD will ensure the following:

- **low power consumption, hence reducing battery weight**
- **removal of visor, eyepiece and CRT equipment will further reduce weight**

The HMD will still be capable of carry out its basic functions like:

- **symbol generation**
- **head protection**
- **weapon aiming**
- **head tracking, etc.**

OLED are built based on organic thin film transistors and they could be made flexible(FOLED) due to their thin (about 0.3mm) and lightweight (about 1.5grams) characteristics.

This project will involve the selection of a suitable trans-missive OLED display and cascading it with a ADC (DE multiplexer). It will also cover the programming and interfacing of a microcontroller for the generation of symbol. After, the developmental integration to a HMD can be carried out.

INTRODUCTION

A helmet-mounted display (HMD) is an aircraft device used to project only important information to the pilot while on flight mission. It is like that of head-up displays (HUD) on a pilot's visor or reticule. A HMD serves as an awareness device to the pilot of the aircraft and environment, and an enhanced scene of object, and in military applications can be used to cue weapons systems, to the direction the head is pointing to. Helmet-mounted sights (HMS) refer to display which allowed cueing equipment system, it is sometimes called Helmet-mounted sight and display (HMSD).

Helmet Optics:

The other important issue is that of providing the pilot with a reference reticule to place over the target. The direction of the line between the reticule and the pilot's eye is the line of sight (LOS) between the aircraft and the intended target. The 'ring and bead' mechanism was adopted from the USA Navy in the seventies, as a strategic solution. But it was always out of focus from the relative aim, due to its nature. And depending on the mounting method, it may move under G load to introduce an unwanted angular error. The next step was to use a flat optical combiner glass, which is based on the same principle as used in world war 2, and project a simple optically collimated reticule, focusing at infinity. This adds some weight to the helmet, but also solves the problem of reticule focusing. A light source such as an LED with a suitable lens package is mounted on top of the helmet can solve this problem. The innovation from optically collimated gunsight into a HUD, with the incandescent bulb replaced by a Cathode Ray Tube (CRT). This integration was introduced during the nineties, thus transforming the HMS into a HMD.

Modern HMDs employ a portable pencil like CRT, embedded into the helmet, and suitable optics to focus the CRT image onto the pilot's visor. Focusing at infinity is often performed by changing visor shape (i.e. curvature) and clever optics. Using this arrangement, a field of view between 10 and 30 degrees can be readily accommodated.

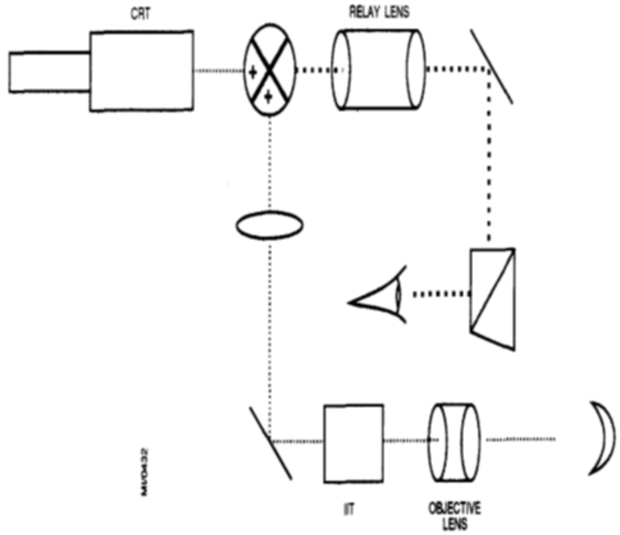
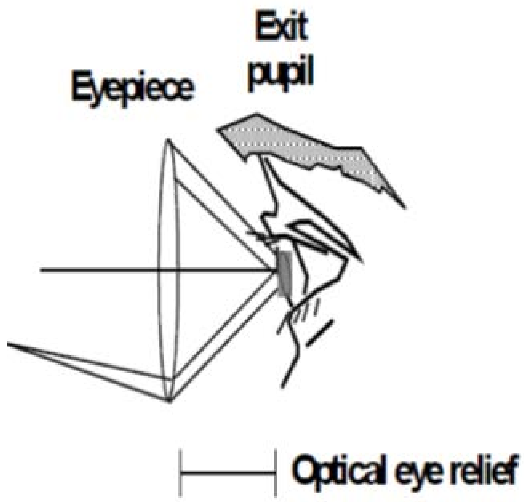


Figure 4optical combination of CRT and intensified

With a CRT, the pilot can be presented with proper symbology, not just circles or crosshairs. seeker acquisition envelope, Missile status data and critical flight data such as load factor, speed, angle of attack (Aoa), Mach number, altitude and horizon may be represented. Cueing boxes produced from radar and RWR/ESM data can be used to mark the position of a target at the limits of visual range. More advanced HMDs can also project raster imagery for the pilot, derived from an external steerable FLIR or an NVG image intensifier tube(s) embedded in the helmet. The most problematic issue now with fighter HMDs is the weight. This technology exists to embed various imaging sensors in the helmet, but this is not particularly compatible with high G maneuvering. In fact, this is one reason why HMDs being used in attack helicopter applications, but are yet to see a wider application in fighters. The other major problematic issue with HMDs, is that the user's eye must always be aligned with the sight, and thus the natural combination of head and eye movement used to track a moving
object must be neglected, and a new "head only" tracking
motion can be learned (used). Since the position of the helmet is what is used to point the missile, pilots would have to develop even greater neck muscles.

The technology is far from its limits. Currently the VR (virtual reality) and AR technologies are much more funded by commercial sources, and are leading to several interesting ideas which are most likely to be utilized in fighter aircraft HMDs. One of such idea, which has been demonstrated in the laboratory, is direct retinal projection. The schemes are based on this model, in which a small mirror is mounted in front of the eye. And a low power laser, or rather a set of three (red, green, blue) lasers are bounced off the mirror directly through the pupil to project a raster scan image directly on the retina. The light beam is scanned by a small piezo-electric actuator that deflects the mirror at a very high speed. This technology claims to provide the potential of 8000 x 4000 pixels for image resolution, while existing technology is limited to that of a VGA class of a computer monitor. Safety is an issue here, since if the mirror actuator fails, could cause a spot burn in your retina. in fact, existing prototypes have elaborated fail sense mechanisms to shut off the laser beams in the event of deflection system failure. There is still some problematic issue with the direct retinal projection. One of which is the existing head only tracking motion problem which is not yet solved. The other is the potential for an eye fatigue, due to the retina being driven with an intense light source, and thus requiring recovery time after use. Another idea from the virtual reality world which could prove to be useful for HMD designs is eye tracking, or sensing the direction the pupil is looking at, relative to the direction of the head. A typical eye tracker will combine a miniature CCD camera and a light emitting diode to illuminate the eye pupil. By measuring the changes in shape and position of the pupil (i.e. an ellipse changing its size and position) it is possible to figure out the direction the eye is looking, with a very reasonable accuracy once calibrated. The Combine measurements obtained from head position and eye position could solve the problems inherent in the already existing HMD technology, since the symbology and weapon aiming are slaved to the direction in which the pilot is looking. Hence retaining existing (instinct) human tracking behavior. The impediment now to the adoption of eye tracking into existing CRT projection HMDs is the limited field of view of the CRT's projection optics. Future HMDs with wider field of view optics will be capable of exploiting eye tracking.

Optical design parameters:
There are a few important descriptive parameters in an HMD optical design. These include:
• Field-of-view (FOV)
• Exit pupil (eye box) size and shape
• Optical eye relief
• Physical eye relief
• Transmission (optical throughput)
• Beam splitter transmission/reflection coefficients (for see-through HMDs)
• Modulation transfer function (MTF)
• Chromatic aberration
• Distortion
• Field curvature
• Magnification
• Ghosting
• Weight (Mass)
• Centre-of-mass (CM)
• Volume (Space required)



Optical eye relief (left) is defined as the distance from the last optical element to the exit pupil,

HMD DESIGN

The range of HMD design types run from a simple projection of symbolic and/or alpha/numeric information overlaying a direct-view real image for day time use, to a head slaved virtual imaging device that could be linked to remote sensors and/or computer generated imagery for day or night use, and of course, anything in between. As the design type increases in complexity, so does the optical design.

Simplest HMD design:

During the Vietnam Era, a Bell Cobra helicopter (AH-1) was developed in Braybrook (1998) with a simple monocular helmet sight (Cobra sight) that could translate an external mounted machinegun using a mechanical head tracker that attached to the top of the helmet. In front of the right eye was a small semi-transparent window that projected a red dot that was like simple commercial red dot reflex sights on some pistols and rifles. The 17 mm diameter combiner was placed outside the helmet visor about 50 mm away from the eye and could be adjusted vertically and horizontally to align with the right eye properly. The size of the projected red dot was only a few milliradians in diameter, and was focused at infinity. The see-through visible transmission of the combining glass (beam-splitter) was very high, and the brightness of the aiming reticule was sufficient to be visible at the sky horizon. Complex modern HMD designs in contrast to the simple design of the Cobra sight is a limited-fielded visor-projection HMD currently being offered by a leading aerospace company having the following characteristics:

• Visor projection optical design
• Focused and aligned at infinity
• Binocular viewing
• Magnetic or electro-optical head tracked
• See-through vision
• FOV – >40°
• Can accommodate a wide range of eye separations
• Sufficient brightness and contrast for both day and night operations
• Incorporates direct-view image intensifiers on the side of the helmet and video links to external
sensors reflected from the visor
• Flight and/or systems symbology can be projected, with unaided daytime see-through vision or at night, overlaying the image intensified or thermal image.

OLED DISPLAY

Introduction:

OLED are solid-state devices, which are composed different layers of thin films of organic molecules, they emit light with the application of current. This Organic LED can provide better and brighter displays for electronic devices and they use less amount of power, as compared to the conventional light emitting diodes that are being used today.

Like an LED, an OLED is a solid-state semiconductor device that is (100 to 500) nm thick or could be about 200 times lesser than a human hair. OLEDs can have either two or three layers of organic material.

OLEDS emit light through a process called electro-phosphorescence. The different types of OLEDs are

• Passive-matrix OLED
• Active-matrix OLED
• Transparent OLED
• Foldable OLED
• Top-emitting OLED
• White OLED Applications

Recently, some small screen devices make use of OLED, such as smart phones, smart watches etc. The research and development in this field is moving faster now more than ever. This could lead to vast future applications in HUD, HMD, automobile dash boards, Motorcycles etc.

Features of OLED:

Organic LED has many intrinsic properties, that gives its unique possibilities of.

• High brightness is achieved at low drive voltages/current densities
• Operating lifetime exceeding 10,000 hours
• Ease of fabrication, since there is no need for crystallization.
• Possible to fabricate on glass and flexible substrates
• Self luminescent so no requirement of backlighting
• Higher brightness
• Low operating and turn-on voltage

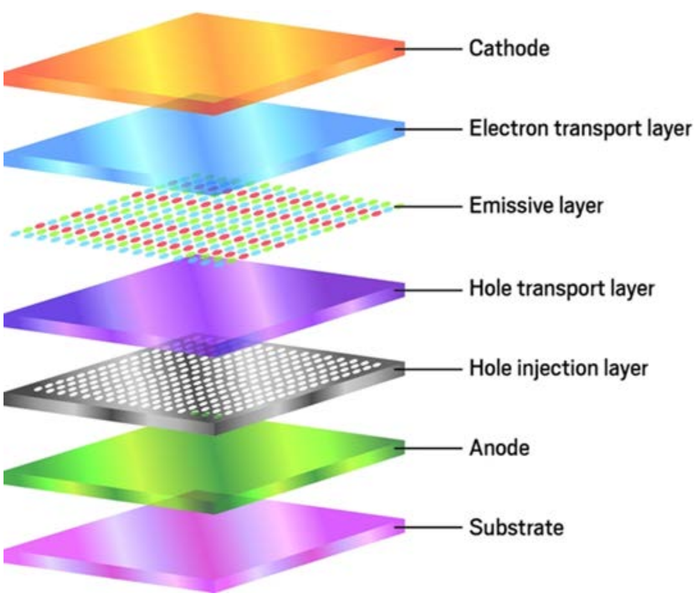
The organic light emitting diode can provide some desirable advantage over the liquid crystal display,

• High contrast
• Low power consumption
• Wide operating temperature range
• Long operating lifetime
• A flexible, thin and lightweight
• Increased brightness

• Faster response time for full motion video Cost effective manufacturability

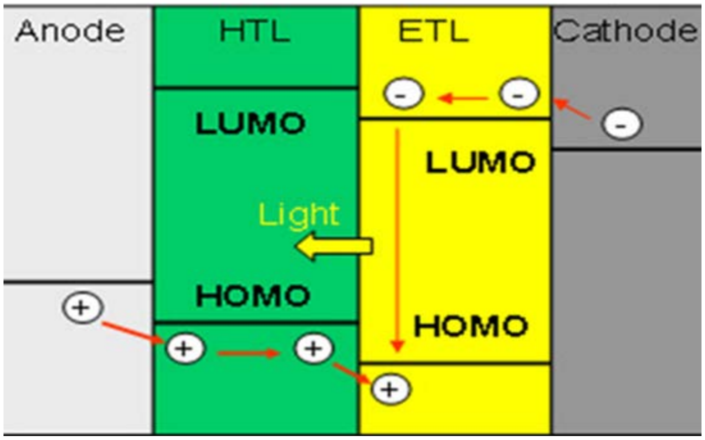
History:
During the early 1950s, the first observations of electroluminescent on organic material was carried out by A. Bernanose and his co-workers at Nancy-Université, France.

OLED COMPONENTS
OLED are a solid-state semiconductor device just like in LEDs, that is about (100 to 500) nanometers thick or about 200 times lesser than a human hair. OLEDs could have organic material of about two to three layers. In recent designs, the third layer usually helps to transport electrons from the cathode to the emissive layer. For our project purpose, we shall be focusing on only two-layer design.
The OLED will consist of the following parts:
• A Substrate (clear plastic, glass, foil): The substrate supports the OLED.
• An Anode (usually transparent): Electrons are removed by the anode (or holes are added) when a current flow through the device.
• The first Organic layer: This is an organic plastic material that transport holes from the anode to the emissive layer. One of such conducting polymer is the polyaniline group.
• An Emissive layer (typically an N-type material): This layer is made of organic plastic molecules (this are different from that of the conducting layer), it recombines electrons with holes, hence photons are emitted inform of light. One such polymer used, is the polyfluorene group.
• A Cathode (For the purpose of our project, we will be spin coating, so as to improve the transparency): Electrons are injected by the cathode when current passes through the device.



OPERATION:
OLEDs uses the process of electro-phosphorescence to emit light, as the manner is in an LED.
The working process is as follows:

1. Applying voltage across the OLED, either thought the use of battery or any other suitable power supply.
2. The electrons will flow from the cathode to the anode, due to the potential difference.
• The cathode yields electrons to the emissive organic layer.
•The anode gains electrons from the conductive organic layer. (This can also be stated as, the addition of electron holes to the conductive layer)
3. At the boundary of the emissive layer and the conductive layer, the electrons will recombine with electron holes.
• When an electron combines with an electron hole, the electron fills the hole (hence its energy level is reduced).
• When this happens, the electron will give up energy in the form of a photon (visible light). This is governed by E=hv
4. The color of the light emitted, depends on the type of organic material used as the emissive layer. Several types of organic materials can be sandwiched, to form colored display.
5. The brightness level of the light depends on the amount of electrical current applied. The more the current, the brighter the light will appear(also known as intensity).

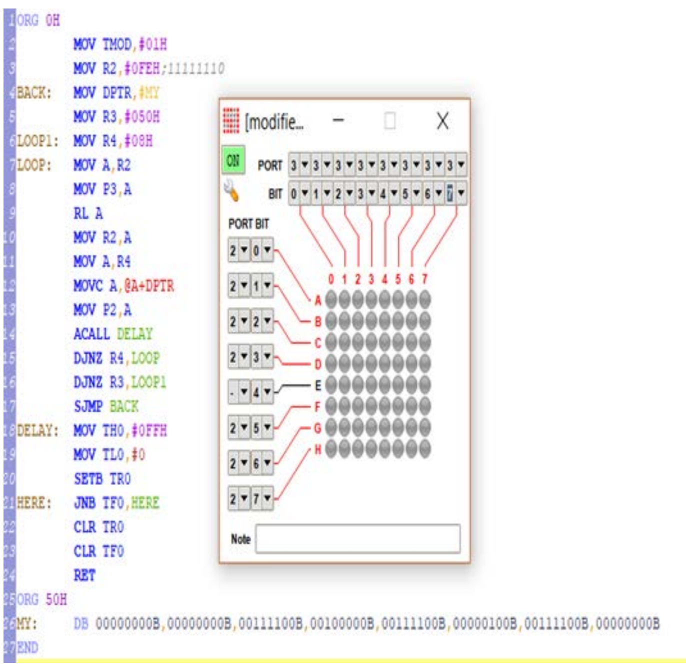


DESIGN OF ELECTRONIC CIRCUIT

Introduction:
The alpha numeric interfacing with 8x8 LED matrix was carried out with the help of 8051 microcontrollers. The objective was to get a single digit display on the LED matrix (i.e. 5).
Software used:
• Proteus
• MCU 8051 IDE
• Symbol generator tool

Program Code:
```
ORG 0H
        MOV TMOD,#01H
        MOV R2,#0FEH;11111110
BACK:   MOV DPTR,#MY
        MOV R3,#050H
LOOP1:  MOV R4,#08H
LOOP:   MOV A,R2
        MOV P3,A
        RL A
```

```
            MOV R2,A
            MOV A,R4
            MOVC A,@A+DPTR
            MOV P2,A
            ACALL DELAY
            DJNZ R4,LOOP
            DJNZ R3,LOOP1
            SJMP BACK
DELAY:  MOV TH0,#0FFH
            MOV TL0,#0
            SETB TR0
HERE:   JNB TF0,HERE
            CLR TR0
            CLR TF0
            RET
ORG 50H
MY:      DB
00000000B,00000000B,00111100B,00100000B,00111100B,00000100B,00111100B,00000000B
END.
```



N/B: The program is done using MCU 8051 IDE software

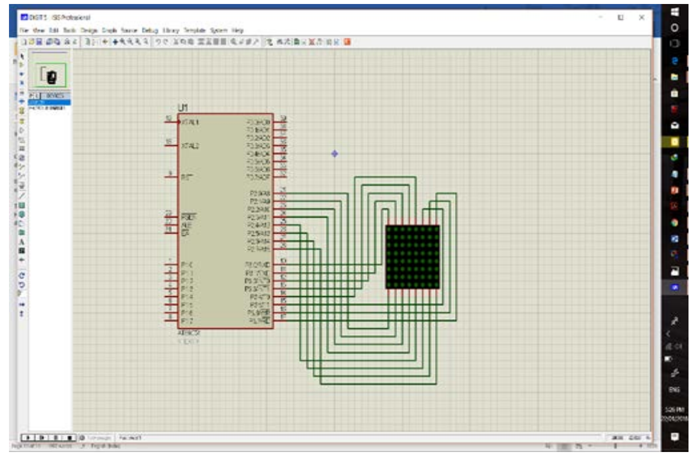
Electronic Circuit Design:
The following component are required for the design of the dot matrix display:

- AT89C51 microcontroller
- 8x8 LED matrix
- Crystal oscillator (for hardware delay)
- 12v power supply

The Proteus software is used for the design of this display.

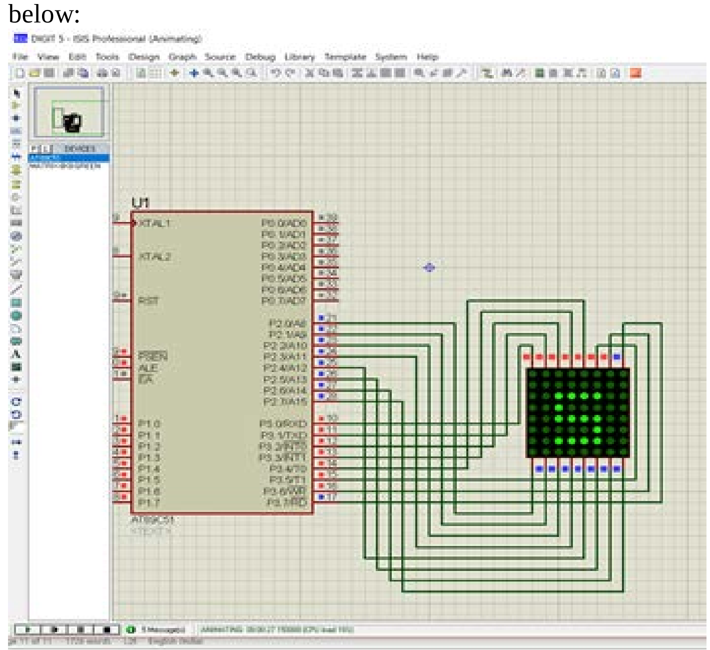
Steps Involved:

1. Open Proteus software and start new project
2. Select the required components as listed above
3. Connect the microcontroller and the LED matrix components as shown below:



4. Import the hex file of the program code into the microcontroller and use clock frequency of 12Mhz

5. Run the simulation and the digit '5' is displayed as shown below:

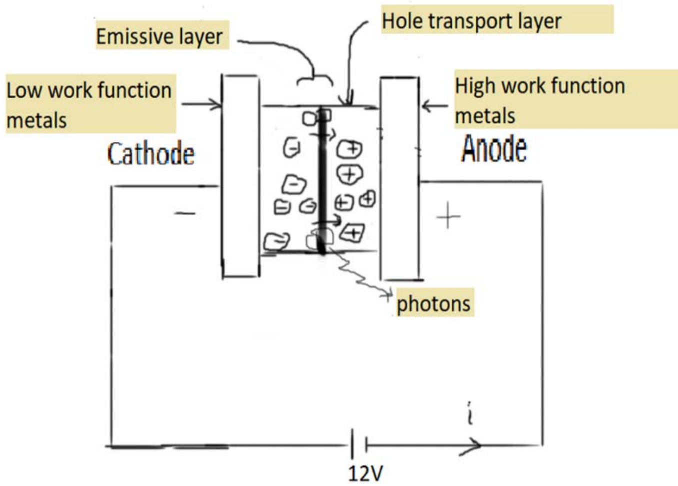


6. Stop the simulation and save the project.

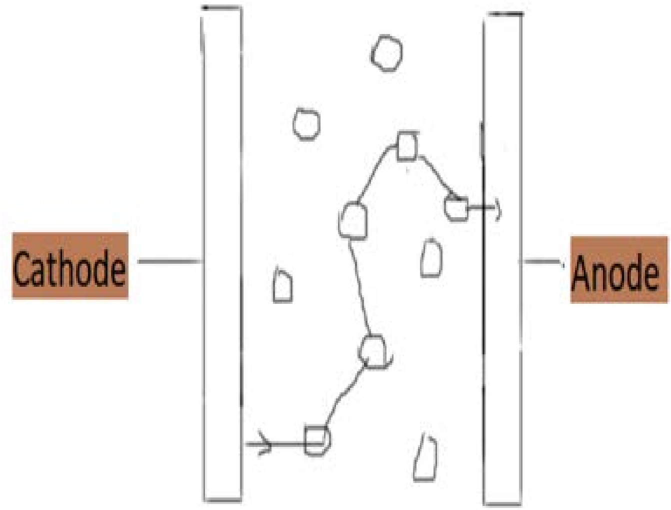
TRANSPARENT OLED

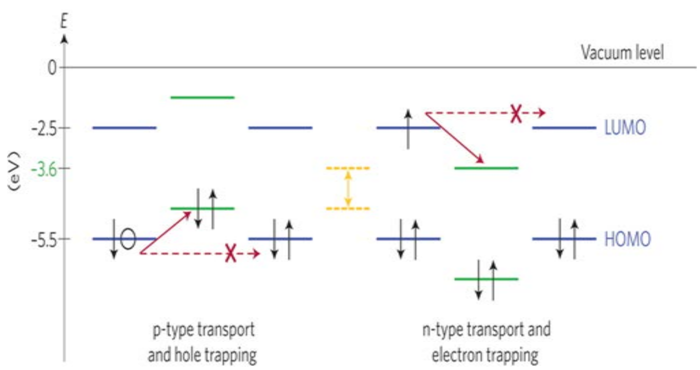
Introduction:
OLEDs are multi-layered light emissive devices that operate of a low applied voltage, typically 3-12Vdc. In an OLEDs simplest form, it is three layer and their most complex, there are five, or more layers. The OLED in this kit is a four-layer device: anode,

hole transport layer (HTL), Red Diamond™ emissive layer, cathode.



The layers are sandwiched between a plastic substrate and an encapsulation layer. Oxygen and water vapor will slowly migrate through the substrate, degrading the devices performance.
The anode is indium tin oxide (ITO) or Aluminum-doped zinc oxide films, that has been pre-deposited onto a PET plastic substrate. The HTL and emissive layers are spin-coated onto the ITO. By altering the spin-coating speed, the thickness of the layers can be altered from 100-500nm. On top of the emissive layer, the cathode is deposited. There are few choices available for the cathode, since it must be a material with a low work-function.
During operation, a voltage (Operating voltage: 2-10V) is applied across the OLED such that the anode is positive with respect to the cathode. Electron charges are made to flow through the device from cathode to the anode, the electrons are injected into the lowest unoccupied molecular orbit (LUMO) of the organic layer at the cathode and are withdrawn from the highest occupied molecular orbit (HOMO) at the anode. This process can also be described as the injection of electron holes into the HOMO. The exitance of Electrostatic forces helps to draw the electrons and holes towards each other. And their recombination forms an exciton (i.e. a bound state of an electron and a hole). This occurs closer to the emissive layer, as it is known that in an organic semiconductor holes are generally in high mobile state than electrons. The decay of this exciton will results in a relaxation of the energy levels of the electron, followed by the emission of radiation (photon) whose frequency is in the spectrum region of visible rays. The frequency of this radiation depends on the band gap of the organic emissive material used, which is the difference in energy between the HOMO and LUMO.
Electrons and holes are known to have half integer spin, so an exciton may either be in a singlet state or in a triplet state. This depends on the recombination spin of the electrons and holes. It is most likely that three triplet excitons will be formed for each singlet exciton. The decay from triplet states is usually spin





forbidden (this is a phosphorescence process). This increases the timescale of the transition and limits the internal efficiency of fluorescent devices. Phosphorescent organic light-emitting diodes make use of spin–orbit interactions to facilitate intersystem crossing between singlet and triplet states, thus obtaining emission from both singlet and triplet states and improving the internal efficiency.

- Trap states
  - Electrons are commonly trapped due to impurities which have empty orbitals, the electrons are usually below -3eV.
  - Holes are not commonly trapped, but could be trapped due to filled orbitals, they are usually above -5eV.

Charge Transport Mechanisms:

This mechanisms shows the theoretical aspect that aims at describing the flow electric current through a given molecular medium, such molecular solids has a characteristic of inter-molecular transport, which is also known as hopping transport

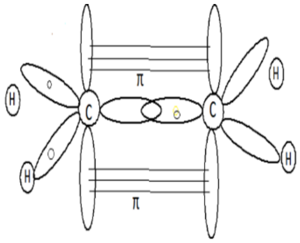


Figure 5 Radom charge path

Material	Size	color	form	Cost (rupees)
2. Poly(ethylenedioxythiophene)/poly(styrenesulfonate) (PEDOT), as a hole injection layer.	5g	blue	liquid	3,500
3. Tris(2,2′-bipyridine)ruthenium(II) hexafluorophosphate, as an emitting layer.	1g	orange	liquid	12,500
5. Indium-tin-oxide (ITO) for an anode.	20x20x1.1 mm	colourless	liquid	1,500 (20x20x1.1 mm)
6. Gallium–Indium eutectic alloy, as cathode.	5g	grey	liquid	7,500 (5g)
Total				25,000

Material Technology:

The following materials from the above table were used for the fabrication of a transparent OLED

1. Indium tin oxide (ITO):

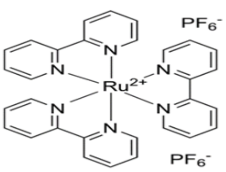
This is a composite of indium, tin and oxygen in different proportions. Based on the oxygen content in the compound, it could either be described as a ceramic or an alloy. Indium tin oxide is usually described as an oxygen-saturated composition with the following by weight mixture 74% In, 18% O2, and 8% Sn. Oxygen-saturated compositions are so common to extent, that an unsaturated composition is termed as oxygen-deficient ITO. It is often transparent and colorless in thin layers, while in bulky form, it could range from yellowish to grey. It sometimes acts as a metal-like mirror within the infrared region of the spectrum.
Indium tin oxide the most commonly used transparent conducting material because of its characteristics of electrical conductivity and optical transparency, as well as the ease of deposition as a thin film on a given medium. The typical problem for all transparent conducting films is the compromise which must be made between conductivity and transparency, since the more the thickness and concentration of charge carriers, will lead to an increase in the material's conductivity, but will also decrease its transparency.

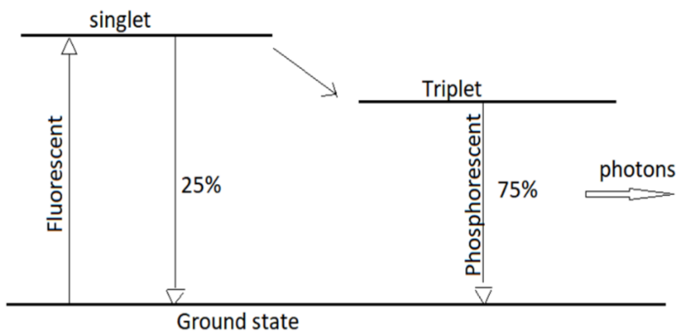
2. PDOT:PSS:

poly(3,4-ethylenedioxythiophene) polystyrene sulfonate is a combination of two ionomers to form a polymer, of which one component in the mixture is made up of sodium polystyrene sulfonate (sulfonated polystyrene). Protons are removed from some part of the sulfonyl groups, so that it carries a negative charge. The poly(3,4-ethylenedioxythiophene) (abbreviated as PEDOT) is second component of conjugated polymer. This carries a positive charge and are based on polythiophene. Together they are used as a macromolecular transparent, conductive polymer with high ductility for various applications. For example in touchscreens, organic light-emitting diodes, flexible organic solar cells and electronic paper to be used alongside the indium tin oxide (ITO), or it could replace it as an electrode. The conductive property of PEDOT:PSS can be effectively improved by a post-treatment with the following chemicals: ethylene glycol, dimethyl sulfoxide (DMSO), salts, zwitterions, co-solvents, acids, alcohols, phenol, germinal diols amphiphilic fluoro-compounds, etc. The conductivity could compared to that of the ITO.

3. Red diamond Ruthenium Complex:

This an organic synthesized emission compound known as Tris(2,2′-bipyridine)ruthenium(II)hexafluorophosphate, Ru(bpy)3(PF6)2. This compounds not only exhibit fluorescent mode but also phosphorescent mode, as shown below:





Due to this it could reach quantum efficiency up to 100%.

4. Gallium–Indium eutectic alloy:

Galinstan is a brand-name and a common name for a liquid metal alloy whose composition is part of a family of eutectic alloys mainly consisting of gallium, indium, and tin. Such eutectic alloys are liquids at room temperature, typically melting at +11 °C (52 °F), while commercial Galinstan melts at −19 °C (−2 °F). An example of a typical eutectic mixture is 68% Ga, 22% In, and 10% Sn (by weight) though proportions vary between 62–95% Ga, 5–22% In, 0–16% Sn (by weight) while remaining eutectic; the exact composition of the commercial product "Galinstan" is not publicly known.

Due to the low toxicity and low reactivity of its component metals, Galinstan finds use as a replacement for many applications that previously employed the toxic liquid mercury or the reactive NaK (sodium–potassium alloy).

Physical properties:

- *Boiling point: > 1300°C*
- *Melting point: −19°C*
- *Vapor pressure: < 1.3-1 KPa (at 500°C)*
- *Density: 6.44 g/cm3 (at 20°C)*
- *Solubility: Insoluble in water or organic solvents*
- *Viscosity: 0.0024 Pa·s (at 20°C)*
- *Thermal conductivity: 16.5 W·m−1·K−1*
- *Electrical conductivity: 3.46×106 S/m (at 20°C)*
- *Surface tension: s = 0.535 N/m (at 20°C)*

Galinstan tends to be "wet" and adhere to many materials, including glass, which limits its use compared to mercury.

Conceptual Model of HMD visor

Deposition Methods:

The different types of deposition methods are as follows :

1. Vacuum Thermal Evaporation
2. Organic vapor phase deposition
3. Inkjet printing:

4. Spin coating:

In this method, the liquid organic material is deposited on a glass substrate in excess. Rotating the substrate at a high velocity will cause the liquid to spread out across the required portion as shown in Fig. 3. This forms a thin transparent layer of the material and will bake as it evaporates. The thickness of the film is determining factor for the thickness of the film is the amount of time expended in rotating the substrate and drying rate of the organic material in use. Films produced this way usually have a problem of inconsistent thickness, as well as poor a surface finish. Because of this non-uniformity, it is not the method of choice, but the cheapest and simplest method of deposition.

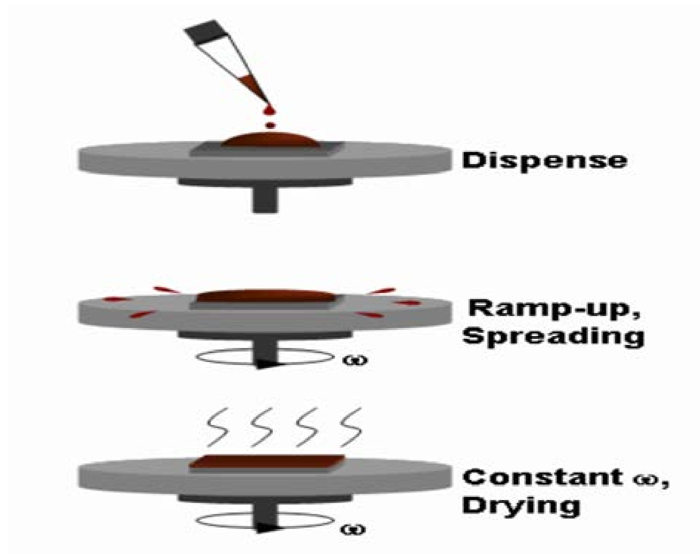


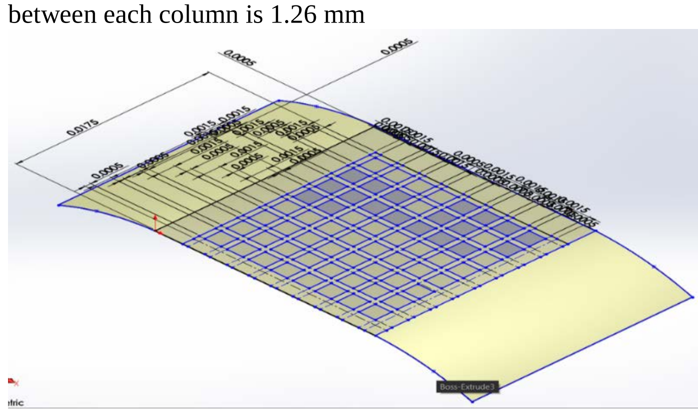
Figure 3

Experimental Modelling The modelling is done using SolidWorks software.

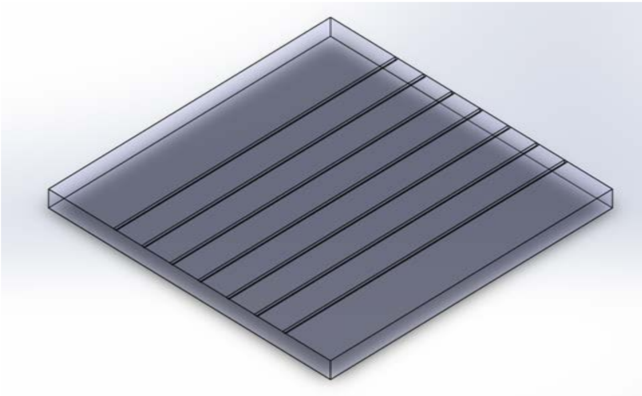
Steps involved:

1. The ITO glass substrate is first modelled using the following dimensions 20x20x1.1 mm for Length, Breath and Thickness, respectively.

• The glass is cleaned and strapped into columns using grooving tool, as shown below for anode connection. Distance between each column is 1.26 mm



2. Modelling of the deposition layer is done for an 8x8 matrix display, using the following dimension shown in the below fig.



• Extrude boss command is used to make surface thickness = 1000nm.

• Extrude cut command is used in creating the matrix pattern, as shown fig.4.

3. The ITO glass part and the pattern deposition layer are assembled and mated together, as shown in fig.4.

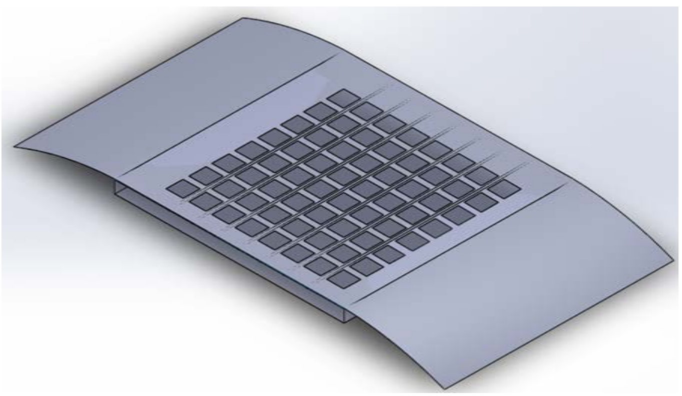


Figure 4

• The (HIL) PDOT:PSS can be applied using pipettes to each pixels and spin coated (deposited), with the help of any motor at 3000 rpm for 60 s and then baked for 30 min at 140 oC . Its appearance is colourless

• The emissive layer red diamond ruthenium complex is also applied to each pixel and spin-deposited at 3000 rpm for 20 s and baked at room temperature. Its appearance is orange in colour, as shown in the fig.6

• The patterning layer is carefully remove

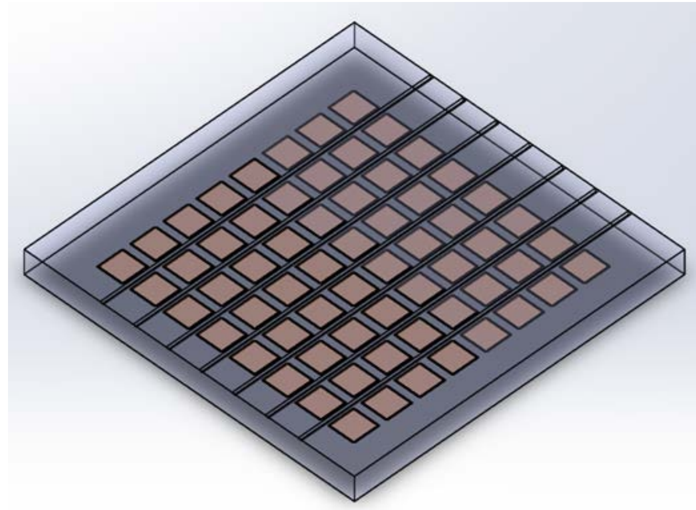
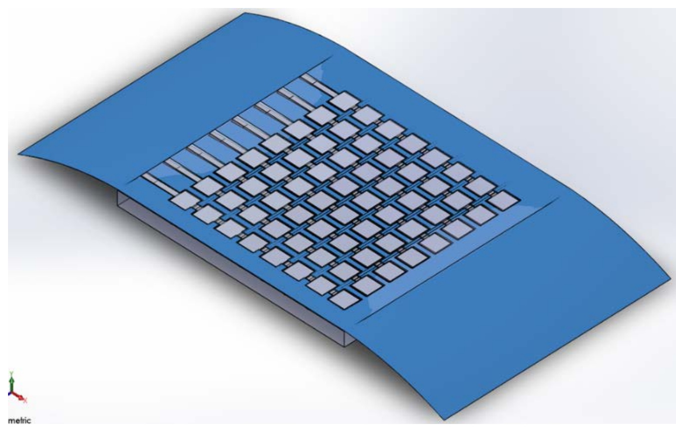
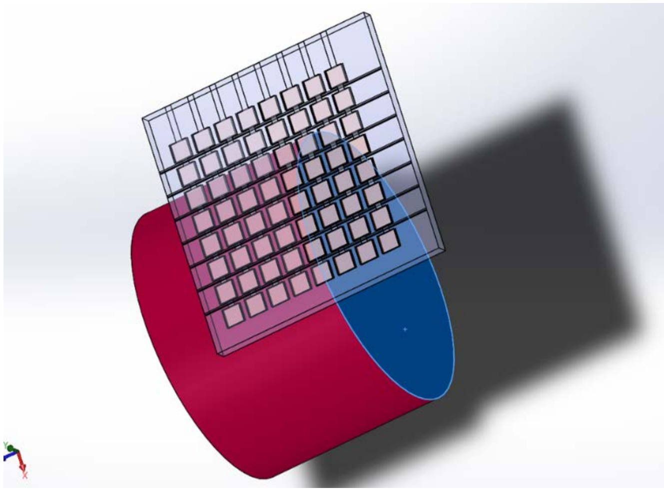


Figure 6

4. The Cathode layer which is the gallium-indium eutectic alloy, is applied to each pixel using a different deposition pattern, with a strap of rows, as shown below:



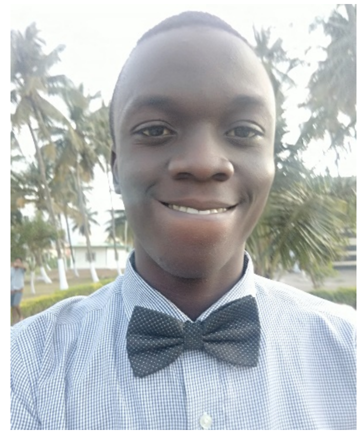
• The alloy is spin-deposited with speed greater than 3000 rpm for 2 min. The figure below shows the overall outcome of the semi-transparent display. An object is placed behind the display to show its transmissivity.



Conclusion:
The alpha numeric interfacing and programming is done using an 8051 MC on a dot 8x8 LED matrix display.
The material selection for a transparent OLED was carried out with its conceptual model, to show what would be the expected outcome of the display. Of which transmittance is would be about (45-70)% (at 500 nm thickness of deposited film layer).

REFRENCES

15. Kevin W. Greeley, "Specialized cockpit displays, symbology, and control mechanization for air vehicle weapons aiming and employment", Proc. SPIE 4362, Cockpit Displays VIII: Displays for Defense Applications, (7 September 2001)
16. J.A. Uphaus, R.B. Bryant, “An innovative tool for designing fault tolerant cockpit display symbology”, Cockpit Integration Div., Wright Lab., Wright-Patterson AFB, OH, USA
17. Kaber DB, Alexander AL, Stelzer EM, Kim SH, Kaufmann K, Hsiang S,”Perceived clutter in advanced cockpit displays: measurement and modeling with experienced pilots”
29.
18. J. Sheats, H. Antoniadis, M. Hueschen, et al, “Organic electroluminescent,” Science 273, 884-888 (1996).
19. C. Tang, and S. VanSlyke, “Organic Electroluminescent Diodes,” Applied Physics Letters 51, 913-915 (1987).
20. Wikipedia contributors, “Spin Coating,” Wikipedia, the free encyclopedia, http://en.wikipedia.org/wiki/Spin_coating (28 November 2006).
21. W. Helfrich and W. G. Schneider, “Recombination radiation in anthracene crystals,” Physical Review Letters 14, 229-231 (1965).
22. Sigma Aldrich contributors, Material science | learning center, Technology spotlight libery, “OLED efabricator” https://www.sigmaaldrich.com/materials-science/learning-center/oled-efabricator.html
23. L.J. Rothberg, and A.J. Lovinger, “Status of and prospects for organic electroluminescence,” Journal of Materials Research 11, 3174-3187 (1996).
24. N. Savage, “Going Organic,” OE Magazine, http://oemagazine.com/FromtheMagazine/nov03/prodtrends.html (28 November 2006).
25. P. E. Burrows, G. Gu, V. Bulovic, Z. Shen, S. R. Forrest, and M. E. Thompson, “Achieving full-color organic light-emitting devices for lightweight, flat-panel displays,” IEEE Transactions on electron devices 44 (8), 1188-1203 (1997).
26. Institute of Chemical Sciences and Engineering, “Device Preparation,” http://ssg.epfl.ch/dev-prep.html coating (28 November 2006).
27. J. Kido, and Y. Iizumi, “Fabrication of highly efficient organic electroluminescent devices,” Applied Physics Letters 73, 2721-2723 (1998).
28. Sugimoto, H. Ochi, S. Fujimura, et al, “Flexible OLED displays using plastic substrates,” IEEE Journal of Selected Topics in Quantum Electronics 10, 107-144

Name	Ezeorah Godswill		Photo 
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SSLC (% of marks and Institution)	80% LOGISS	HSC/ Diploma (As above)	80%
B. Tech (up to VII Sem)	8.24 (CGPA)	M. Tech (up to III Sem)	-
Future plan	Higher Studies	Area of Interest	Dynamics and Propulsion
In-plant training	-	IAESTE	-
Placement status (Company Name)	Nil	Mentor name	Mr. Prashant kumar
Project title (UG)	PROOF OF CONCEPT FOR TRANSPARENT HMD USING OLED		
Project Guide	Lt. Musica SR., Assistant Professor, KITS.		
Conference attended	1) ICSIE 2018, Chennai.		
Gate score/ TANCET/ GRE/TOFEL	-		
Academic and sports awards received	-		
Software skills developed	ANSYS, CATIA, AUTOCAD, MATLAB, LabVIEW, MCU-IDE, ATMEL STUDIO, SOLIDWORKS, PROTEUS, KEIL, etc.		
Previous experience if any	Nil		
Date	10/4/2018	Signature	